

Theory of Algebraic Topology 3

July 29, 2026

Lecture notes with worked examples in algebraic topology

Contents

1 The covering homomorphism $SU(2) \rightarrow SO(3)$

Explanation. One of the most important examples of a covering group is the map

$$SU(2) \rightarrow SO(3).$$

Its meaning is that every element of $SU(2)$ determines a rotation of $\mathbb{R}^3$, and two elements U and $-U$ determine the same rotation. Thus $SU(2)$ is a double cover of $SO(3)$.

Step 1: the two groups.

The group $SO(3)$ is the group of rotations of $\mathbb{R}^3$:

$$SO(3) = \{A \in M_3(\mathbb{R}) : A^T A = I, \det A = 1\}.$$

Its elements are exactly the usual real 3×3 rotation matrices.

The group $SU(2)$ is

$$SU(2) = \{U \in M_2(\mathbb{C}) : U^* U = I, \det U = 1\}.$$

Every element of $SU(2)$ has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

If we write

$$\alpha = a + bi, \quad \beta = c + di,$$

then the condition becomes

$$a^2 + b^2 + c^2 + d^2 = 1.$$

So as a topological space,

$$SU(2) \cong S^3.$$

Step 2: identify $\mathbb{R}^3$ with a space of matrices.

To connect $SU(2)$ with rotations of $\mathbb{R}^3$, we identify $\mathbb{R}^3$ with the space of traceless Hermitian 2×2 complex matrices.

To a vector

$$(x, y, z) \in \mathbb{R}^3$$

we associate the matrix

$$X = \begin{pmatrix} z & x + iy \\ x - iy & -z \end{pmatrix}.$$

This matrix is Hermitian, because $X^* = X$, and traceless, because

$$\text{tr}(X) = z + (-z) = 0.$$

Conversely, every traceless Hermitian 2×2 matrix has this form. Hence $\mathbb{R}^3$ is identified with the real vector space

$$V = \left\{ \begin{pmatrix} z & x + iy \\ x - iy & -z \end{pmatrix} : x, y, z \in \mathbb{R} \right\}.$$

Step 3: define the action of $SU(2)$.

Let $U \in SU(2)$. Define a map on V by conjugation:

$$X \mapsto UXU^{-1}.$$

Since U is unitary, $U^{-1} = U^*$. One checks that if X is Hermitian and traceless, then UXU^{-1} is again Hermitian and traceless. So this gives a linear map

$$V \rightarrow V.$$

Under the identification $V \cong \mathbb{R}^3$, this becomes a linear map

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

Thus each $U \in SU(2)$ defines a transformation of $\mathbb{R}^3$. Denote it by $\varphi(U)$. This gives a map

$$\varphi : SU(2) \rightarrow GL(3, \mathbb{R}).$$

Step 4: why $\varphi(U)$ lies in $SO(3)$.

The determinant of the matrix X above is

$$\det(X) = -(x^2 + y^2 + z^2).$$

So the quantity

$$x^2 + y^2 + z^2$$

is preserved by conjugation, because determinant is preserved:

$$\det(UXU^{-1}) = \det(X).$$

Hence $\varphi(U)$ preserves Euclidean length on $\mathbb{R}^3$. Therefore

$$\varphi(U) \in O(3).$$

Since $SU(2)$ is connected and φ is continuous, the image of $SU(2)$ lies in the connected component of the identity in $O(3)$, namely $SO(3)$. So in fact

$$\varphi : SU(2) \rightarrow SO(3).$$

Step 5: φ is a group homomorphism.

For $U, V \in SU(2)$ and $X \in V$,

$$\varphi(UV)(X) = (UV)X(UV)^{-1} = U(VXV^{-1})U^{-1}.$$

Thus

$$\varphi(UV) = \varphi(U)\varphi(V).$$

So φ is a group homomorphism.

Step 6: the kernel.

If $U = \pm I$, then

$$UXU^{-1} = X$$

for every $X \in V$. Hence

$$\{\pm I\} \subseteq \text{Ker}(\varphi).$$

Conversely, suppose $U \in \text{Ker}(\varphi)$. Then

$$UXU^{-1} = X \quad \text{for all } X \in V.$$

So U commutes with every traceless Hermitian matrix. These matrices span the full matrix algebra $M_2(\mathbb{C})$, so U must commute with every 2×2 complex matrix. Therefore U must be a scalar matrix:

$$U = \lambda I.$$

Since $U \in SU(2)$, we have

$$|\lambda| = 1, \quad \det(U) = \lambda^2 = 1.$$

Hence

$$\lambda = \pm 1.$$

So

$$\text{Ker}(\varphi) = \{\pm I\}.$$

Step 7: surjectivity.

Every rotation in $SO(3)$ is a rotation about some axis by some angle. One can show that every such rotation arises from some $U \in SU(2)$. Therefore φ is surjective.

Hence

$$SO(3) \cong SU(2)/\{\pm I\}.$$

Since the kernel has two elements, the map

$$\varphi : SU(2) \rightarrow SO(3)$$

is a 2-sheeted covering homomorphism.

Step 8: why this is the universal covering.

We already know that

$$SU(2) \cong S^3.$$

Since S^3 is simply connected, $SU(2)$ is simply connected. Therefore the covering

$$SU(2) \rightarrow SO(3)$$

is the universal covering of $SO(3)$.

Thus $SU(2)$ is the universal covering group of $SO(3)$.

Step 9: relation with $\mathbb{RP}^3$.

Because $SU(2) \cong S^3$ and the kernel is $\{\pm I\}$, we get

$$SO(3) \cong SU(2)/\{\pm I\} \cong S^3/\{\pm 1\} \cong \mathbb{RP}^3.$$

So the map

$$SU(2) \rightarrow SO(3)$$

is exactly the same topological phenomenon as the antipodal quotient

$$S^3 \rightarrow \mathbb{RP}^3.$$

Geometric meaning. Upstairs, each element of $SU(2) \cong S^3$ represents a rotation. Downstairs, in $SO(3)$, the two opposite points U and $-U$ represent the same rotation. Thus every rotation of $\mathbb{R}^3$ has exactly two lifts to $SU(2)$.

So the picture is

$$SU(2) \cong S^3 \longrightarrow SO(3) \cong \mathbb{RP}^3,$$

where the map is 2-to-1.

Quaternion interpretation. There is another very useful description. The group $SU(2)$ may be identified with the group of unit quaternions

$$q = a + bi + cj + dk, \quad a^2 + b^2 + c^2 + d^2 = 1.$$

This again gives the 3-sphere S^3 .

The space $\mathbb{R}^3$ is identified with the imaginary quaternions

$$bi + cj + dk.$$

A unit quaternion q acts on $\mathbb{R}^3$ by

$$v \longmapsto qvq^{-1}.$$

This action preserves lengths and gives a rotation of $\mathbb{R}^3$. Thus we again obtain a homomorphism

$$S^3 \rightarrow SO(3).$$

The kernel is

$$\{\pm 1\},$$

because q and $-q$ act in the same way:

$$(-q)v(-q)^{-1} = qvq^{-1}.$$

So this quaternion description is exactly the same covering homomorphism as

$$SU(2) \rightarrow SO(3).$$

Summary. The map

$$SU(2) \rightarrow SO(3)$$

is defined by conjugation on traceless Hermitian 2×2 matrices, or equivalently by conjugation of imaginary quaternions. It is a surjective homomorphism with kernel

$$\{\pm I\}.$$

Therefore

$$SO(3) \cong SU(2)/\{\pm I\},$$

and since $SU(2) \cong S^3$ is simply connected, it is the universal covering group of $SO(3)$.

2 Axis-angle description in $SO(3)$ and its lift to $SU(2)$

Explanation. The relation between $SO(3)$ and $SU(2)$ can be understood very concretely through the axis-angle description of rotations.

An element of $SO(3)$ is an actual rotation of $\mathbb{R}^3$. An element of $SU(2)$ is an *upstairs representative* of that same rotation. It encodes the same axis, but the formula uses half the rotation angle.

1. Rotations in $SO(3)$.

Every nontrivial element $R \in SO(3)$ is a rotation of $\mathbb{R}^3$ by some angle

$$\theta \in (0, \pi]$$

about some axis

$$u \in S^2.$$

Thus the geometric data downstairs is

$$(\text{axis } u, \text{ angle } \theta).$$

If $\theta = \pi$, then the axes u and $-u$ determine the same rotation. Thus in this special case the axis is determined only up to sign.

How to extract the axis from a rotation matrix.

Let

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} \in SO(3).$$

The rotation axis is the direction left fixed by R . Therefore the axis vector u satisfies

$$Ru = u,$$

equivalently,

$$(R - I)u = 0.$$

So u is any nonzero vector in the kernel of $R - I$, normalized to have length 1.

This is the most conceptual formula:

axis u is any unit vector in $\ker(R - I)$.

Angle from the trace.

The angle θ is determined by

$$\cos \theta = \frac{\text{tr}(R) - 1}{2}.$$

Explicit axis formula when $\theta \neq 0, \pi$.

If $\theta \notin \{0, \pi\}$, then the axis can be recovered from the skew-symmetric part of R :

$$u = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix},$$

up to sign.

Equivalently,

$$u \propto \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}.$$

This comes from Rodrigues' formula, which gives

$$R - R^T = 2 \sin \theta \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}.$$

Special cases.

- If $\theta = 0$, then $R = I$, so every axis works.
- If $\theta = \pi$, then the formula involving $\sin \theta$ breaks down. In that case one solves

$$(R - I)u = 0$$

directly.

Example: rotation around the z -axis.

Consider

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$r_{32} - r_{23} = 0, \quad r_{13} - r_{31} = 0, \quad r_{21} - r_{12} = 2 \sin \theta.$$

So the axis is

$$u = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

2. Upstairs in $SU(2)$.

An element of $SU(2)$ corresponds to the same geometric rotation, but upstairs it is written using the *half-angle* $\theta/2$.

Using the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and writing

$$u = (u_1, u_2, u_3) \in S^2,$$

the element of $SU(2)$ corresponding to rotation by angle θ around axis u is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (u_1 \sigma_1 + u_2 \sigma_2 + u_3 \sigma_3).$$

Thus the axis upstairs is still the same axis u , but the angle appearing in the formula is $\theta/2$, not θ .

Why the half-angle appears.

If one replaces θ by $\theta + 2\pi$, then downstairs in $SO(3)$ this is the same rotation, but upstairs the corresponding element changes sign:

$$\cos \frac{\theta + 2\pi}{2} = \cos \left(\frac{\theta}{2} + \pi \right) = -\cos \frac{\theta}{2},$$

and similarly

$$\sin \frac{\theta + 2\pi}{2} = -\sin \frac{\theta}{2}.$$

Hence the element U is replaced by $-U$.

So the two elements

$$U \quad \text{and} \quad -U$$

of $SU(2)$ correspond to the same element of $SO(3)$. This is exactly why the map

$$SU(2) \rightarrow SO(3)$$

is 2-to-1.

Interpretation.

Thus:

- in $SO(3)$, a nontrivial element is literally a rotation of $\mathbb{R}^3$ by angle θ around axis u ;
- in $SU(2)$, an element is an upstairs representative of that same rotation;
- it remembers the same axis u , but it uses the half-angle $\theta/2$.

So the correspondence is:

$$\begin{aligned} \text{downstairs in } SO(3) : & \quad (\text{axis } u, \text{ angle } \theta), \\ \text{upstairs in } SU(2) : & \quad \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (u \cdot \sigma). \end{aligned}$$

3. Quaternion form.

The same picture can be written more geometrically with quaternions. Identify $SU(2)$ with the group of unit quaternions

$$q = a + bi + cj + dk, \quad a^2 + b^2 + c^2 + d^2 = 1.$$

This is again the 3-sphere S^3 .

Let

$$u = (u_1, u_2, u_3) \in S^2.$$

Then the unit quaternion representing rotation by angle θ about axis u is

$$q = \cos \frac{\theta}{2} + \sin \frac{\theta}{2}(u_1 i + u_2 j + u_3 k).$$

The space $\mathbb{R}^3$ is identified with the imaginary quaternions

$$u_1 i + u_2 j + u_3 k.$$

The corresponding rotation is given by

$$v \longmapsto qvq^{-1}.$$

Again:

- the axis is u ,
- the geometric rotation downstairs is by angle θ ,
- the upstairs formula uses $\theta/2$.

Also,

$$(-q)v(-q)^{-1} = qvq^{-1},$$

so q and $-q$ determine the same rotation. Thus the kernel of the covering map is

$$\{\pm 1\}.$$

4. A 2π -rotation and a 4π -rotation.

This half-angle phenomenon explains a famous fact.

A full 2π -rotation in $SO(3)$ is the identity rotation. But upstairs in $SU(2)$, a 2π -rotation corresponds to

$$U = -I,$$

not to I . Only a 4π -rotation returns to I upstairs.

So:

$$2\pi \text{ downstairs} = \text{identity in } SO(3),$$

but

$$2\pi \text{ upstairs} = -I \in SU(2),$$

and

$$4\pi \text{ upstairs} = I.$$

This is one of the clearest manifestations of the fact that

$$SU(2) \rightarrow SO(3)$$

is a double cover.

Summary. The group $SO(3)$ consists of actual rotations of $\mathbb{R}^3$, each described by an axis u and an angle θ . The group $SU(2)$ consists of upstairs representatives of these same rotations. It keeps the same axis, but the formula uses the half-angle $\theta/2$. Two opposite elements U and $-U$ in $SU(2)$ represent the same rotation downstairs in $SO(3)$.

Thus the geometric picture is

$$SU(2) \cong S^3 \longrightarrow SO(3) \cong \mathbb{RP}^3,$$

with kernel

$$\{\pm I\},$$

and this is the universal covering map of $SO(3)$.

2.1 From a rotation matrix $R \in SO(3)$ to its axis v , angle θ , and lift $U \in SU(2)$

Explanation. Starting from a rotation matrix

$$R \in SO(3),$$

one can extract:

- the rotation axis $v \in S^2$,
- the rotation angle $\theta \in [0, \pi]$,
- and then construct the corresponding lift $U \in SU(2)$.

Thus the chain is

$$R \in SO(3) \implies (\theta, v) \implies U \in SU(2).$$

Here R is the actual rotation downstairs, v is its axis in $\mathbb{R}^3$, θ is its angle, and U is one of the two elements upstairs mapping to R .

1. Extracting the axis v from R .

The axis of the rotation is the direction fixed by R . Therefore v satisfies

$$Rv = v,$$

equivalently,

$$(R - I)v = 0.$$

So the most conceptual formula is:

the axis v is any unit vector in $\ker(R - I)$.

After solving for a nonzero vector in $\ker(R - I)$, one normalizes it to length 1.

2. Extracting the angle θ .

The rotation angle is determined by the trace formula

$$\boxed{\cos \theta = \frac{\operatorname{tr}(R) - 1}{2}}, \quad \theta \in [0, \pi].$$

3. Explicit axis formula when $\theta \neq 0, \pi$.

Let

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}.$$

If $\theta \notin \{0, \pi\}$, then the axis can be extracted from the skew-symmetric part of R :

$$\boxed{v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

Equivalently,

$$\boxed{v \propto \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

This comes from Rodrigues' formula, which implies

$$R - R^T = 2 \sin \theta \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}.$$

4. Special cases.

- If $\theta = 0$, then $R = I$, so this is the identity rotation and every axis works.
- If $\theta = \pi$, then the formula involving $1/(2 \sin \theta)$ breaks down, because $\sin \pi = 0$. In that case one solves

$$(R - I)v = 0$$

directly.

5. Constructing the lift $U \in SU(2)$.

Once one has the axis

$$v = (v_1, v_2, v_3) \in S^2$$

and the angle θ , the corresponding element upstairs in $SU(2)$ is built using the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then the lift is

$$\boxed{U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3)}.$$

This is one lift of R . The other lift is

$$\boxed{-U.}$$

The two matrices U and $-U$ determine the same element of $SO(3)$.

6. Expanded matrix form of the lift.

Since

$$v_1\sigma_1 + v_2\sigma_2 + v_3\sigma_3 = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix},$$

we obtain

$$\boxed{U = \begin{pmatrix} \cos \frac{\theta}{2} - iv_3 \sin \frac{\theta}{2} & (-iv_1 - v_2) \sin \frac{\theta}{2} \\ (-iv_1 + v_2) \sin \frac{\theta}{2} & \cos \frac{\theta}{2} + iv_3 \sin \frac{\theta}{2} \end{pmatrix}.}$$

Again, both U and $-U$ map to the same rotation $R \in SO(3)$.

7. Practical algorithm.

Given a rotation matrix $R \in SO(3)$, proceed as follows:

1. Compute the angle from

$$\cos \theta = \frac{\text{tr}(R) - 1}{2}.$$

2. Find the axis v :

- always valid:

$$(R - I)v = 0, \quad \|v\| = 1;$$

- or, if $\theta \neq 0, \pi$,

$$v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}.$$

3. Form the lift

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma).$$

Then the two lifts of R are

$$\boxed{U \quad \text{and} \quad -U.}$$

8. Example: rotation about the z -axis.

Consider the matrix

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$(R - I) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0,$$

so the axis is

$$v = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The corresponding lift is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_3 = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

The other lift is

$$-U = \begin{pmatrix} -e^{-i\theta/2} & 0 \\ 0 & -e^{i\theta/2} \end{pmatrix}.$$

Both correspond to the same rotation matrix $R \in SO(3)$.

9. Summary.

From a rotation matrix $R \in SO(3)$, one extracts the angle by

$$\cos \theta = \frac{\text{tr}(R) - 1}{2},$$

and the axis by

$$v \text{ is any unit vector in } \ker(R - I).$$

Then the corresponding elements of $SU(2)$ are

$$\pm \left(\cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma) \right).$$

Thus every rotation matrix downstairs determines exactly two points upstairs in $SU(2)$, differing by sign.

2.2 From a rotation matrix $R \in SO(3)$ to its axis v , angle θ , and lift $U \in SU(2)$

Explanation. Starting from a rotation matrix

$$R \in SO(3),$$

one can extract:

- the rotation axis $v \in S^2$,
- the rotation angle $\theta \in [0, \pi]$,
- and then construct the corresponding lift $U \in SU(2)$.

Thus the chain is

$$R \in SO(3) \implies (\theta, v) \implies U \in SU(2).$$

Here R is the actual rotation downstairs, v is its axis in $\mathbb{R}^3$, θ is its angle, and U is one of the two elements upstairs mapping to R .

1. Extracting the axis v from R .

The axis of the rotation is the direction fixed by R . Therefore v satisfies

$$Rv = v,$$

equivalently,

$$(R - I)v = 0.$$

So the most conceptual formula is:

$$\boxed{\text{the axis } v \text{ is any unit vector in } \ker(R - I).}$$

After solving for a nonzero vector in $\ker(R - I)$, one normalizes it to length 1.

2. Extracting the angle θ .

The rotation angle is determined by the trace formula

$$\boxed{\cos \theta = \frac{\operatorname{tr}(R) - 1}{2}}, \quad \theta \in [0, \pi].$$

3. Explicit axis formula when $\theta \neq 0, \pi$.

Let

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}.$$

If $\theta \notin \{0, \pi\}$, then the axis can be extracted from the skew-symmetric part of R :

$$\boxed{v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

Equivalently,

$$\boxed{v \propto \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

This comes from Rodrigues' formula, which implies

$$R - R^T = 2 \sin \theta \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}.$$

4. Special cases.

- If $\theta = 0$, then $R = I$, so this is the identity rotation and every axis works.
- If $\theta = \pi$, then the formula involving $1/(2 \sin \theta)$ breaks down, because $\sin \pi = 0$. In that case one solves

$$(R - I)v = 0$$

directly.

5. The Pauli matrices and the meaning of $v \cdot \sigma$.

The Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

When one writes

$$\sigma = (\sigma_1, \sigma_2, \sigma_3),$$

this does *not* mean one single matrix. It means the ordered triple of Pauli matrices.

If

$$v = (v_1, v_2, v_3) \in \mathbb{R}^3,$$

then

$$v \cdot \sigma$$

means the dot product

$$v \cdot \sigma = v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3.$$

Expanding this gives

$$v \cdot \sigma = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

6. Constructing the lift $U \in SU(2)$.

Once one has the axis

$$v = (v_1, v_2, v_3) \in S^2$$

and the angle θ , the corresponding element upstairs in $SU(2)$ is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma).$$

This is one lift of R . The other lift is

$$-U.$$

The two matrices U and $-U$ determine the same element of $SO(3)$.

7. Expanded matrix form of the lift.

Using the explicit expression for $v \cdot \sigma$, we obtain

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

Hence

$$U = \begin{pmatrix} \cos \frac{\theta}{2} - iv_3 \sin \frac{\theta}{2} & (-iv_1 - v_2) \sin \frac{\theta}{2} \\ (-iv_1 + v_2) \sin \frac{\theta}{2} & \cos \frac{\theta}{2} + iv_3 \sin \frac{\theta}{2} \end{pmatrix}.$$

Again, both U and $-U$ map to the same rotation $R \in SO(3)$.

8. Practical algorithm.

Given a rotation matrix $R \in SO(3)$, proceed as follows:

1. Compute the angle from

$$\cos \theta = \frac{\operatorname{tr}(R) - 1}{2}.$$

2. Find the axis v :

- always valid:

$$(R - I)v = 0, \quad \|v\| = 1;$$

- or, if $\theta \neq 0, \pi$,

$$v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}.$$

3. Form the lift

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma).$$

Then the two lifts of R are

$$\boxed{U \quad \text{and} \quad -U.}$$

9. Example: rotation about the z -axis.

Consider the matrix

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$(R - I) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0,$$

so the axis is

$$v = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Now

$$v \cdot \sigma = \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore the lift is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_3 = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

The other lift is

$$-U = \begin{pmatrix} -e^{-i\theta/2} & 0 \\ 0 & -e^{i\theta/2} \end{pmatrix}.$$

Both correspond to the same rotation matrix $R \in SO(3)$.

10. Summary.

From a rotation matrix $R \in SO(3)$, one extracts the angle by

$$\cos \theta = \frac{\operatorname{tr}(R) - 1}{2},$$

and the axis by

$$v \text{ is any unit vector in } \ker(R - I).$$

Then the corresponding elements of $SU(2)$ are

$$\pm \left(\cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma) \right).$$

Here

$$v \cdot \sigma = v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3 = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

Thus every rotation matrix downstairs determines exactly two points upstairs in $SU(2)$, differing by sign.

2.3 From a rotation matrix $R \in SO(3)$ to its axis v , angle θ , and lift $U \in SU(2)$

Explanation. Starting from a rotation matrix

$$R \in SO(3),$$

one can extract:

- the rotation axis $v \in S^2$,
- the rotation angle $\theta \in [0, \pi]$,
- and then construct the corresponding lift $U \in SU(2)$.

Thus the chain is

$$R \in SO(3) \implies (\theta, v) \implies U \in SU(2).$$

Here R is the actual rotation downstairs, v is its axis in $\mathbb{R}^3$, θ is its angle, and U is one of the two elements upstairs mapping to R .

1. Extracting the axis v from R .

The axis of the rotation is the direction fixed by R . Therefore v satisfies

$$Rv = v,$$

equivalently,

$$(R - I)v = 0.$$

So the most conceptual formula is:

$$\text{the axis } v \text{ is any unit vector in } \ker(R - I).$$

After solving for a nonzero vector in $\ker(R - I)$, one normalizes it to length 1.

2. Extracting the angle θ .

The rotation angle is determined by the trace formula

$$\boxed{\cos \theta = \frac{\operatorname{tr}(R) - 1}{2}}, \quad \theta \in [0, \pi].$$

3. Explicit axis formula when $\theta \neq 0, \pi$.

Let

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}.$$

If $\theta \notin \{0, \pi\}$, then the axis can be extracted from the skew-symmetric part of R :

$$\boxed{v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

Equivalently,

$$\boxed{v \propto \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}}.$$

This comes from Rodrigues' formula, which implies

$$R - R^T = 2 \sin \theta \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}.$$

4. Special cases.

- If $\theta = 0$, then $R = I$, so this is the identity rotation and every axis works.
- If $\theta = \pi$, then the formula involving $1/(2 \sin \theta)$ breaks down, because $\sin \pi = 0$. In that case one solves

$$(R - I)v = 0$$

directly.

5. The Pauli matrices and the meaning of $v \cdot \sigma$.

The Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

When one writes

$$\sigma = (\sigma_1, \sigma_2, \sigma_3),$$

this does *not* mean one single matrix. It means the ordered triple of Pauli matrices.

If

$$v = (v_1, v_2, v_3) \in \mathbb{R}^3,$$

then

$$v \cdot \sigma$$

means the dot product

$$v \cdot \sigma = v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3.$$

Expanding this gives

$$v \cdot \sigma = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

6. Constructing the lift $U \in SU(2)$.

Once one has the axis

$$v = (v_1, v_2, v_3) \in S^2$$

and the angle θ , the corresponding element upstairs in $SU(2)$ is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma).$$

This is one lift of R . The other lift is

$$-U.$$

The two matrices U and $-U$ determine the same element of $SO(3)$.

7. Expanded matrix form of the lift.

Using the explicit expression for $v \cdot \sigma$, we obtain

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

Hence

$$U = \begin{pmatrix} \cos \frac{\theta}{2} - iv_3 \sin \frac{\theta}{2} & (-iv_1 - v_2) \sin \frac{\theta}{2} \\ (-iv_1 + v_2) \sin \frac{\theta}{2} & \cos \frac{\theta}{2} + iv_3 \sin \frac{\theta}{2} \end{pmatrix}.$$

Again, both U and $-U$ map to the same rotation $R \in SO(3)$.

8. Practical algorithm.

Given a rotation matrix $R \in SO(3)$, proceed as follows:

1. Compute the angle from

$$\cos \theta = \frac{\text{tr}(R) - 1}{2}.$$

2. Find the axis v :

- always valid:

$$(R - I)v = 0, \quad \|v\| = 1;$$

- or, if $\theta \neq 0, \pi$,

$$v = \frac{1}{2 \sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix}.$$

3. Form the lift

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma).$$

Then the two lifts of R are

$$\boxed{U \quad \text{and} \quad -U.}$$

9. Example: rotation about the z -axis.

Consider the matrix

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$(R - I) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0,$$

so the axis is

$$v = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Now

$$v \cdot \sigma = \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore the lift is

$$U = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_3 = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

The other lift is

$$-U = \begin{pmatrix} -e^{-i\theta/2} & 0 \\ 0 & -e^{i\theta/2} \end{pmatrix}.$$

Both correspond to the same rotation matrix $R \in SO(3)$.

10. Summary.

From a rotation matrix $R \in SO(3)$, one extracts the angle by

$$\boxed{\cos \theta = \frac{\operatorname{tr}(R) - 1}{2}},$$

and the axis by

$$\boxed{v \text{ is any unit vector in } \ker(R - I).}$$

Then the corresponding elements of $SU(2)$ are

$$\pm \left(\cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (v \cdot \sigma) \right).$$

Here

$$v \cdot \sigma = v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3 = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}.$$

Thus every rotation matrix downstairs determines exactly two points upstairs in $SU(2)$, differing by sign.

3 Problem 1 Universal cover of $S^1 \vee S^2$

Problem statement. Recall that for based spaces (X, x_0) , (Y, y_0) , the wedge product $X \vee Y$ is the quotient of the space

$$X \sqcup Y$$

by the relation generated by $x_0 \sim y_0$. What is the universal cover of $S^1 \vee S^2$? Draw a picture.

Solution. Since S^2 is simply connected, wedging it to S^1 does not change the fundamental group:

$$\pi_1(S^1 \vee S^2) \cong \pi_1(S^1) \cong \mathbb{Z}.$$

So the universal cover should be a simply connected space on which $\mathbb{Z}$ acts by deck transformations.

Start from the universal cover of S^1 , namely

$$\mathbb{R} \rightarrow S^1.$$

Now in $S^1 \vee S^2$, the sphere S^2 is attached at the wedge point. In the universal cover, every lift of that wedge point gets its own copy of S^2 . The lifts of the wedge point are the integers in $\mathbb{R}$. Therefore the universal cover is:

a copy of $\mathbb{R}$ with one S^2 attached at each integer point.

Equivalently,

$$\widetilde{S^1 \vee S^2} \cong (\mathbb{R}) \cup \bigcup_{n \in \mathbb{Z}} S_n^2,$$

where each S_n^2 is attached to $\mathbb{R}$ at the point $n \in \mathbb{R}$.

This space is simply connected: the line $\mathbb{R}$ is simply connected, and each attached sphere is simply connected, so no new fundamental group appears.

A picture is:

$$\dots - \bullet - \bullet - \bullet - \bullet - \dots$$

with a 2-sphere attached at each bullet. The bullets correspond to the integers in $\mathbb{R}$.

So the universal cover of $S^1 \vee S^2$ is an infinite chain of copies of S^2 , indexed by $\mathbb{Z}$, joined by a line.

4 Problem 2 Quotients by free properly discontinuous actions and odd-dimensional lens spaces

Problem statement. Let X be a Hausdorff space, and G a group acting on X by homeomorphisms, freely (i.e. if $g \in G$ satisfies $g \cdot x = x$ for some $x \in X$, then $g = e$) and properly discontinuously (i.e. each $x \in X$ has an open neighbourhood $U \ni x$ such that

$$\{g \in G \mid g(U) \cap U \neq \emptyset\}$$

is finite).

- (i) Show that the quotient map $X \rightarrow X/G$ is a covering map.
- (ii) Deduce that if X is simply connected and locally path-connected then for any point $[x] \in X/G$ we have an isomorphism of groups

$$\pi_1(X/G, [x]) \cong G.$$

- (iii) Hence show that for $n \geq 2$ odd and any $m \geq 2$ there is a space Y with fundamental group $\mathbb{Z}/m$ and universal cover S^n . [Hint: Consider S^n as the unit sphere in $\mathbb{C}^k$.]

Solution.

(i) The quotient map is a covering map. Let $p : X \rightarrow X/G$ be the quotient map, and fix $x \in X$.

By proper discontinuity, there exists an open neighbourhood $U \ni x$ such that

$$F := \{g \in G \mid g(U) \cap U \neq \emptyset\}$$

is finite.

For each $g \in F \setminus \{e\}$, we have $gx \neq x$ because the action is free. Since X is Hausdorff, we can choose an open neighbourhood $U_g \subseteq U$ of x such that

$$U_g \cap g(U_g) = \emptyset.$$

Now set

$$W := \bigcap_{g \in F \setminus \{e\}} U_g.$$

Then W is an open neighbourhood of x , and for every $g \neq e$,

$$g(W) \cap W = \emptyset.$$

Indeed, if $g(W) \cap W \neq \emptyset$, then since $W \subseteq U$ we get $g(U) \cap U \neq \emptyset$, hence $g \in F$, contradicting the construction of W .

Now consider $p(W) \subseteq X/G$. We claim that

$$p^{-1}(p(W)) = \bigsqcup_{g \in G} g(W).$$

This is clear: a point $y \in X$ lies in $p^{-1}(p(W))$ exactly when $p(y) = p(w)$ for some $w \in W$, i.e. $y = g(w)$ for some $g \in G$, so $y \in g(W)$.

Also the sets $g(W)$ are pairwise disjoint. Indeed, if

$$g_1(W) \cap g_2(W) \neq \emptyset,$$

then

$$W \cap (g_1^{-1}g_2)(W) \neq \emptyset,$$

so $g_1^{-1}g_2 = e$, hence $g_1 = g_2$.

Finally, for each $g \in G$, the restriction

$$p|_{g(W)} : g(W) \rightarrow p(W)$$

is a homeomorphism, because p is injective on $g(W)$ and surjective onto $p(W)$, and p is a quotient map.

Thus $p^{-1}(p(W))$ is a disjoint union of open sets each mapped homeomorphically onto $p(W)$. Therefore $p : X \rightarrow X/G$ is a covering map.

(ii) Computation of the fundamental group. Assume now that X is simply connected and locally path-connected.

By part (i), the quotient map

$$p : X \rightarrow X/G$$

is a covering map. Since X is simply connected, this covering is the universal covering of X/G .

The group G acts on X by homeomorphisms, and each element $g \in G$ preserves fibres of p , so each g is a deck transformation of the universal cover. Conversely, if two points $x, y \in X$ lie in the same fibre of p , then $y = gx$ for a unique $g \in G$. Hence the deck transformation group is exactly G .

For a universal cover of a connected, locally path-connected space, the deck transformation group is isomorphic to the fundamental group of the base. Therefore

$$\pi_1(X/G, [x]) \cong G.$$

(iii) A space with fundamental group $\mathbb{Z}/m$ and universal cover S^n . Let $n \geq 2$ be odd. Then

$$n = 2k - 1$$

for some $k \geq 2$, and we may view

$$S^n = S^{2k-1} = \left\{ (z_1, \dots, z_k) \in \mathbb{C}^k : \sum_{j=1}^k |z_j|^2 = 1 \right\}.$$

Let $\mu_m \subset \mathbb{C}^\times$ be the group of m -th roots of unity:

$$\mu_m = \{\zeta \in \mathbb{C} : \zeta^m = 1\} \cong \mathbb{Z}/m.$$

Define an action of μ_m on S^{2k-1} by

$$\zeta \cdot (z_1, \dots, z_k) = (\zeta z_1, \dots, \zeta z_k).$$

This action is free. Indeed, if

$$\zeta \cdot (z_1, \dots, z_k) = (z_1, \dots, z_k),$$

then for every coordinate j ,

$$\zeta z_j = z_j.$$

Since the point lies on the sphere, not all z_j are zero, so for some j we have $z_j \neq 0$, hence $\zeta = 1$.

Since μ_m is finite, the action is automatically properly discontinuous.

Therefore, by part (ii), the quotient

$$Y := S^{2k-1}/\mu_m$$

has

$$\pi_1(Y) \cong \mu_m \cong \mathbb{Z}/m,$$

and its universal cover is $S^{2k-1} = S^n$.

This quotient is a standard lens space.

5 Problem 3 Isomorphisms of groups given by presentations

Problem statement.

(i) Show that the groups

$$G = \langle a, b \mid a \rangle \quad \text{and} \quad H = \langle t \mid \rangle$$

are isomorphic.

(ii) Show that the groups

$$G = \langle a, b \mid ab^{-3}, ba^{-2} \rangle \quad \text{and} \quad H = \langle t \mid t^5 \rangle$$

are isomorphic.

(iii) Show that the groups

$$G = \langle a, b \mid a^3b^{-2} \rangle \quad \text{and} \quad H = \langle x, y \mid xyxy^{-1}x^{-1}y^{-1} \rangle$$

are isomorphic. Show that this group is non-abelian and infinite.

Solution.

(i) In

$$G = \langle a, b \mid a \rangle,$$

the relation $a = 1$ kills the generator a . Hence G is generated only by b , with no further relation:

$$G \cong \langle b \mid \rangle \cong \mathbb{Z}.$$

Also

$$H = \langle t \mid \rangle \cong \mathbb{Z}.$$

Therefore

$$G \cong H.$$

(ii) The relations are

$$ab^{-3} = 1, \quad ba^{-2} = 1,$$

so

$$a = b^3, \quad b = a^2.$$

Substituting $a = b^3$ into $b = a^2$, we get

$$b = (b^3)^2 = b^6,$$

hence

$$b^5 = 1.$$

Also $a = b^3$, so every element of G is a power of b . Thus

$$G \cong \langle b \mid b^5 \rangle \cong \mathbb{Z}/5.$$

But

$$H = \langle t \mid t^5 \rangle \cong \mathbb{Z}/5.$$

So

$$G \cong H.$$

An explicit isomorphism is given by

$$\varphi : G \rightarrow H, \quad \varphi(b) = t, \quad \varphi(a) = t^3.$$

(iii) First rewrite the relation in H :

$$xyxy^{-1}x^{-1}y^{-1} = 1.$$

This is equivalent to

$$xyx = yxy.$$

Indeed,

$$xyxy^{-1}x^{-1}y^{-1} = 1 \iff xyxy^{-1}x^{-1} = y \iff xyxy^{-1} = yx \iff xyx = yxy.$$

So

$$H = \langle x, y \mid xyx = yxy \rangle,$$

the standard braid-group-on-three-strands presentation.

Now define

$$\varphi : G \rightarrow H$$

by

$$\varphi(a) = xy, \quad \varphi(b) = xyx.$$

We check the defining relation of G :

$$\varphi(a)^3 = (xy)^3 = xyxyxy.$$

Using $xyx = yxy$,

$$(xy)^3 = (xyx)(yxy) = (xyx)^2 = \varphi(b)^2.$$

Hence

$$\varphi(a^3b^{-2}) = 1,$$

so φ is a well-defined homomorphism.

Now define

$$\psi : H \rightarrow G$$

by

$$\psi(x) = a^{-1}b, \quad \psi(y) = b^{-1}a^2.$$

Then

$$\psi(x)\psi(y) = a^{-1}bb^{-1}a^2 = a,$$

so

$$\psi(x)\psi(y)\psi(x) = a(a^{-1}b) = b.$$

Also

$$\psi(y)\psi(x)\psi(y) = \psi(y)a = b^{-1}a^2a = b^{-1}a^3 = b^{-1}b^2 = b.$$

Thus

$$\psi(x)\psi(y)\psi(x) = \psi(y)\psi(x)\psi(y),$$

so ψ respects the relation $xyx = yxy$, hence is well-defined.

Now compute:

$$\psi(\varphi(a)) = \psi(xy) = a, \quad \psi(\varphi(b)) = \psi(xyx) = b,$$

and also

$$\varphi(\psi(x)) = \varphi(a^{-1}b) = (xy)^{-1}(xyx) = x,$$

$$\varphi(\psi(y)) = \varphi(b^{-1}a^2) = (xyx)^{-1}(xy)^2 = y.$$

Therefore φ and ψ are inverse isomorphisms, so

$$G \cong H.$$

The group is non-abelian. Consider the homomorphism

$$G \rightarrow S_3$$

defined by

$$a \mapsto (123), \quad b \mapsto (12).$$

Then

$$a^3 = e, \quad b^2 = e,$$

so the relation $a^3 = b^2$ is satisfied. Hence this is a well-defined homomorphism. Its image contains a 3-cycle and a transposition, so the image is all of S_3 , which is non-abelian. Therefore G , and hence H , is non-abelian.

The group is infinite. Define a homomorphism

$$G \rightarrow \mathbb{Z}$$

by

$$a \mapsto 2, \quad b \mapsto 3.$$

This is well-defined because the relation $a^3 = b^2$ becomes

$$3 \cdot 2 = 2 \cdot 3,$$

which is true in $\mathbb{Z}$. The image contains 2 and 3, hence contains $\gcd(2, 3) = 1$, so it is all of $\mathbb{Z}$. Thus G surjects onto an infinite group, and therefore G itself is infinite.

So the group in part (iii) is isomorphic to B_3 , non-abelian, and infinite.

6 Theory accompanying Problems 1–3

6.1 Wedge sums, universal covers, and lifting simply connected pieces

Wedge sums. Let (X, x_0) and (Y, y_0) be based topological spaces. Their *wedge sum* is

$$X \vee Y := (X \sqcup Y)/(x_0 \sim y_0).$$

Thus $X \vee Y$ is obtained by taking the disjoint union of X and Y and identifying the chosen basepoints. Geometrically, this means that the two spaces are glued together at exactly one point.

Typical examples are:

$$S^1 \vee S^1 \quad (\text{the figure eight}), \quad S^1 \vee S^2 \quad (\text{a circle and a sphere meeting at one point}).$$

Covering spaces. A map

$$p : \widetilde{X} \rightarrow X$$

is a *covering map* if every point $x \in X$ has an open neighbourhood $U \subseteq X$ such that

$$p^{-1}(U) = \bigsqcup_{\alpha \in A} U_\alpha,$$

where each $U_\alpha \subseteq \widetilde{X}$ is open and the restriction

$$p|_{U_\alpha} : U_\alpha \rightarrow U$$

is a homeomorphism.

A covering

$$p : \widetilde{X} \rightarrow X$$

is called a *universal cover* if $\widetilde{X}$ is simply connected.

For connected, locally path-connected, semilocally simply connected spaces, the universal cover exists and is unique up to isomorphism.

Fundamental group of a wedge. A standard consequence of van Kampen's theorem is that, for suitably nice based spaces,

$$\pi_1(X \vee Y) \cong \pi_1(X) * \pi_1(Y),$$

the free product of the two fundamental groups.

In particular,

$$\pi_1(S^1 \vee S^2) \cong \pi_1(S^1) * \pi_1(S^2) \cong \mathbb{Z} * 1 \cong \mathbb{Z}.$$

So in the wedge $S^1 \vee S^2$, all nontrivial loop information comes from the S^1 -part; the S^2 -part contributes no fundamental group because S^2 is simply connected.

A useful lifting principle. Suppose a space is obtained by attaching a simply connected space at one point to a space that carries the nontrivial fundamental group. Then, in the universal cover, the simply connected piece appears unchanged over each lift of the attaching point.

This principle explains the universal cover of

$$S^1 \vee S^2.$$

Indeed, the universal cover of S^1 is $\mathbb{R}$, and the wedge point lifts to the integer points of $\mathbb{R}$. Therefore the universal cover of $S^1 \vee S^2$ is the space obtained by attaching one copy of S^2 at every integer point of $\mathbb{R}$.

Schematically:

$$\widetilde{S^1 \vee S^2} \cong \mathbb{R} \cup \bigcup_{n \in \mathbb{Z}} S_n^2,$$

where the sphere S_n^2 is attached at the point $n \in \mathbb{R}$.

Deck transformations in this example. Since

$$\pi_1(S^1 \vee S^2) \cong \mathbb{Z},$$

the deck transformation group of the universal cover is $\mathbb{Z}$. It acts by translations

$$x \longmapsto x + k, \quad k \in \mathbb{Z},$$

on the line $\mathbb{R}$, and this shifts the attached spheres accordingly. Thus the universal cover has a repeating $\mathbb{Z}$ -pattern.

This gives a useful consistency check: whenever one proposes a universal cover, its deck transformation group should match the fundamental group of the base.

6.2 Free properly discontinuous actions and quotient coverings

Group actions by homeomorphisms. Let G be a group and X a topological space. An action of G on X by homeomorphisms means that each $g \in G$ determines a homeomorphism

$$X \rightarrow X, \quad x \mapsto g \cdot x,$$

such that

$$e \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x)$$

for all $g, h \in G$ and $x \in X$.

For $x \in X$, the *orbit* of x is

$$Gx = \{g \cdot x : g \in G\}.$$

The quotient space X/G is the set of orbits equipped with the quotient topology.

Free actions. The action is called *free* if

$$g \cdot x = x \implies g = e$$

for every $x \in X$. In other words, no nontrivial element of G fixes any point of X .

This condition is essential for quotient-covering constructions: if a nontrivial group element fixed a point, then the quotient could develop singular behaviour rather than locally looking like a disjoint union of identical sheets.

Properly discontinuous actions. The action is called *properly discontinuous* if every point $x \in X$ has an open neighbourhood $U \ni x$ such that

$$\{g \in G : g(U) \cap U \neq \emptyset\}$$

is finite.

Intuitively, this means that near any point, only finitely many translates of a small neighbourhood can return and meet it. In many standard situations one can shrink U further so that

$$g(U) \cap U = \emptyset \quad \text{for all } g \neq e.$$

This is exactly the local separation needed to obtain an evenly covered neighbourhood in the quotient.

Why the quotient map is a covering map. Let G act freely and properly discontinuously on a Hausdorff space X , and let

$$p : X \rightarrow X/G$$

be the quotient map.

Fix $x \in X$. By proper discontinuity, there exists a neighbourhood $U \ni x$ such that only finitely many translates meet U . Since the action is free and X is Hausdorff, we can shrink U to an open neighbourhood $W \ni x$ so that

$$g(W) \cap W = \emptyset \quad \text{for all } g \neq e.$$

Then

$$p^{-1}(p(W)) = \bigsqcup_{g \in G} g(W),$$

and each restriction

$$p|_{g(W)} : g(W) \rightarrow p(W)$$

is a homeomorphism.

Thus $p(W)$ is evenly covered, and therefore

$$p : X \rightarrow X/G$$

is a covering map.

Universal covers from quotient actions. Suppose now that X is simply connected and locally path-connected, and that G acts freely and properly discontinuously on X . Then the quotient map

$$X \rightarrow X/G$$

is not only a covering map but in fact the universal covering map. Hence the deck transformation group of this covering is G , and therefore

$$\pi_1(X/G) \cong G.$$

This gives one of the most important geometric constructions of spaces with prescribed fundamental group:

start with a simply connected space X , let a group G act freely and properly discontinuously on X , then the quotient X/G has fundamental group G .

Odd-dimensional spheres and cyclic quotients. Let $n \geq 2$ be odd. Then

$$n = 2k - 1$$

for some $k \geq 2$, and we may view

$$S^{2k-1} = \left\{ (z_1, \dots, z_k) \in \mathbb{C}^k : \sum_{j=1}^k |z_j|^2 = 1 \right\}.$$

Let

$$\mu_m = \{\zeta \in \mathbb{C} : \zeta^m = 1\} \cong \mathbb{Z}/m$$

be the group of m -th roots of unity. Define an action of μ_m on S^{2k-1} by

$$\zeta \cdot (z_1, \dots, z_k) = (\zeta z_1, \dots, \zeta z_k).$$

This action is free: if

$$\zeta \cdot (z_1, \dots, z_k) = (z_1, \dots, z_k),$$

then, since not all z_j are zero, some coordinate satisfies $z_j \neq 0$, so $\zeta z_j = z_j$, hence $\zeta = 1$.

Since μ_m is finite, the action is properly discontinuous. Therefore

$$S^{2k-1} \rightarrow S^{2k-1}/\mu_m$$

is a universal covering map, and

$$\pi_1(S^{2k-1}/\mu_m) \cong \mu_m \cong \mathbb{Z}/m.$$

These quotients are the classical lens spaces.

Why odd dimension appears here. The scalar multiplication construction

$$(z_1, \dots, z_k) \mapsto (\zeta z_1, \dots, \zeta z_k)$$

naturally acts on the unit sphere in $\mathbb{C}^k$, which is

$$S^{2k-1}.$$

Thus the sphere dimension coming from this standard construction is necessarily odd.

6.3 Groups given by presentations

What a presentation means. A group presentation

$$\langle a, b, \dots \mid r_1, r_2, \dots \rangle$$

means that we begin with the free group on the generators $a, b, \dots$, and then impose the relations

$$r_1 = 1, \quad r_2 = 1, \quad \dots$$

Equivalently, we quotient the free group by the normal subgroup generated by the relators $r_1, r_2, \dots$.

So a presentation specifies a group by saying:

these are the generators, and these are the only relations we force among them.

Eliminating generators using relations. A common technique in presentation problems is to solve some relations for one generator in terms of others.

For example,

$$\langle a, b \mid a \rangle$$

means that $a = 1$. So the generator a disappears, and the group becomes

$$\langle b \mid \rangle \cong \mathbb{Z}.$$

Similarly, in

$$\langle a, b \mid ab^{-3}, ba^{-2} \rangle,$$

the relators say

$$a = b^3, \quad b = a^2.$$

Substituting one into the other reduces the presentation to a group on one generator. This is one of the standard strategies: rewrite the relations as explicit algebraic identities and simplify step by step.

Two standard ways to prove two presented groups are isomorphic.

Method 1: simplify both presentations. Reduce each presentation until both become obviously the same familiar group.

Method 2: construct explicit homomorphisms. Define a homomorphism on generators, check that the defining relators are sent to the identity, and then construct an inverse map.

The second method is especially important for nonabelian groups, where simple elimination is often not enough. This is exactly what happens in the comparison

$$\langle a, b \mid a^3 b^{-2} \rangle \quad \text{and} \quad \langle x, y \mid xyx = yxy \rangle.$$

The braid relation. The relation

$$xyx = yxy$$

is called the *braid relation*. The group

$$\langle x, y \mid xyx = yxy \rangle$$

is the braid group B_3 on three strands.

This is a fundamental example of a nonabelian infinite group. It appears in topology, algebra, geometry, mapping class groups, knot theory, and many other places.

A classical fact is that

$$\langle a, b \mid a^3 = b^2 \rangle$$

gives another presentation of a group isomorphic to B_3 . Thus the relation

$$a^3 b^{-2} = 1$$

is closely connected with the braid relation.

Proving nonabelianity via quotients. A very useful general principle is:

If a group surjects onto a nonabelian group, then it is itself nonabelian.

So to prove that a presented group is nonabelian, it is often enough to construct a surjective homomorphism onto a known nonabelian group such as S_3 .

For example, if one can define a homomorphism

$$G \rightarrow S_3$$

whose image contains both a transposition and a 3-cycle, then the image is all of S_3 , and G cannot be abelian.

Proving infinitude via quotients. Another useful principle is:

If a group surjects onto an infinite group, then it is infinite.

Thus, to prove that a presented group is infinite, one often tries to construct a surjective homomorphism onto $\mathbb{Z}$.

For instance, if the defining relation is

$$a^3 = b^2,$$

then sending

$$a \mapsto 2, \quad b \mapsto 3$$

gives a homomorphism to $\mathbb{Z}$, because in additive notation the relation becomes

$$3 \cdot 2 = 2 \cdot 3,$$

which is true. Since 2 and 3 generate $\mathbb{Z}$, the map is surjective, and so the original group is infinite.

A standard workflow for presentation problems. When working with a presented group, a useful sequence of questions is:

1. Can the relators be rewritten more transparently?
2. Can one eliminate a generator?
3. Does the presentation reduce to a known group?
4. Can one build explicit homomorphisms to compare with another presentation?
5. Can one detect nonabelianity by surjecting onto a known nonabelian group?
6. Can one detect infinitude by surjecting onto $\mathbb{Z}$?

This workflow is especially effective for small presentations, and it is exactly the kind of reasoning used in Problems 3(i)–(iii).

6.4 A compact problem-solving guide for these topics

For universal-cover problems. A good procedure is:

1. compute or identify the fundamental group downstairs,
2. recognize which part of the space produces the nontrivial loops,
3. lift that part first,
4. then attach simply connected pieces over each lift of the attaching set,
5. check that the resulting deck transformation group has the correct form.

For wedge spaces, van Kampen's theorem is usually the main algebraic tool.

For quotient-by-action problems. A good checklist is:

1. verify that the action is by homeomorphisms,

2. verify that the action is free,
3. verify that the action is properly discontinuous,
4. deduce that the quotient map is a covering map,
5. if the original space is simply connected, conclude that it is the universal cover,
6. identify the deck group with the acting group and conclude

$$\pi_1(X/G) \cong G.$$

For group-presentation problems. A good checklist is:

1. rewrite relators in a simpler algebraic form,
2. eliminate generators whenever possible,
3. compare with standard presentations,
4. construct explicit homomorphisms both ways if an isomorphism is claimed,
5. use quotients to test whether the group is abelian or nonabelian,
6. use quotients to test whether the group is finite or infinite.

These principles tie together the three problems above: one uses covering-space theory, one uses quotient actions, and one uses combinatorial group theory, but in each case the method is to identify the correct structure and then exploit its universal property.

7 Problem 4 The fundamental group of $S^1 \vee S^1$ and nonexistence of a retraction

Problem statement. Use Seifert–van Kampen to show that the fundamental group of $S^1 \vee S^1$ is a free group on two generators. Deduce that the inclusion

$$i : S^1 \vee S^1 = (S^1 \times \{*\}) \cup (\{*\} \times S^1) \hookrightarrow S^1 \times S^1$$

does not admit a retraction.

Solution. Let

$$X = S^1 \vee S^1.$$

We regard X as the union of two circles meeting in one point x_0 . Choose open neighbourhoods $U, V \subseteq X$ such that

$$U \simeq S^1, \quad V \simeq S^1, \quad U \cap V \text{ is contractible,}$$

and

$$X = U \cup V.$$

Then Seifert–van Kampen gives

$$\pi_1(X, x_0) \cong \pi_1(U, x_0) * \pi_1(V, x_0).$$

Since

$$\pi_1(U, x_0) \cong \mathbb{Z}, \quad \pi_1(V, x_0) \cong \mathbb{Z},$$

we obtain

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z}.$$

Thus $\pi_1(S^1 \vee S^1, x_0)$ is the free group on two generators. If a and b denote the two characteristic loops, we may write

$$\pi_1(S^1 \vee S^1, x_0) \cong F_2 = \langle a, b \rangle.$$

Now consider the inclusion

$$i : S^1 \vee S^1 \hookrightarrow S^1 \times S^1.$$

The torus has fundamental group

$$\pi_1(S^1 \times S^1) \cong \mathbb{Z}^2,$$

and the induced homomorphism

$$i_* : \pi_1(S^1 \vee S^1, x_0) \rightarrow \pi_1(S^1 \times S^1, i(x_0))$$

sends the two generators a, b to the two standard generators of $\mathbb{Z}^2$. Hence

$$i_* : F_2 \rightarrow \mathbb{Z}^2$$

is the abelianization map. In particular,

$$i_*([a, b]) = 0,$$

so i_* is not injective.

Assume for contradiction that there exists a retraction

$$r : S^1 \times S^1 \rightarrow S^1 \vee S^1$$

such that

$$r \circ i = \text{id}_{S^1 \vee S^1}.$$

Then on fundamental groups we would have

$$r_* \circ i_* = \text{id}_{\pi_1(S^1 \vee S^1, x_0)}.$$

Therefore i_* would have to be injective, which is impossible.

Hence the inclusion

$$i : S^1 \vee S^1 \hookrightarrow S^1 \times S^1$$

does not admit a retraction.

8 Problem 5 Connected degree 2 and degree 3 coverings of $S^1 \vee S^1$

Problem statement. A covering space is called normal if it corresponds to a normal subgroup. Draw pictures of all the connected degree 2 covering spaces of $S^1 \vee S^1$. Show that they are all normal coverings. Now do the same thing for the connected degree 3 covering spaces of $S^1 \vee S^1$. Which of them are normal coverings?

Solution. Let

$$X = S^1 \vee S^1, \quad \pi_1(X) \cong F_2 = \langle a, b \rangle.$$

A connected degree n covering of X is described by a transitive permutation representation

$$\rho : F_2 \rightarrow S_n.$$

Equivalently, it is determined by a pair of permutations

$$\sigma_a = \rho(a), \quad \sigma_b = \rho(b)$$

in S_n such that the subgroup generated by σ_a, σ_b acts transitively on $\{1, \dots, n\}$. Two such coverings are isomorphic if and only if the two pairs are simultaneously conjugate in S_n .

So we classify connected coverings by classifying transitive pairs (σ_a, σ_b) up to simultaneous conjugacy.

Degree 2. Here

$$S_2 = \{1, \tau\}, \quad \tau = (12).$$

For the action to be transitive, at least one of σ_a, σ_b must be τ . Thus, up to isomorphism, there are exactly three connected degree 2 coverings.

(1) $\sigma_a = \tau, \sigma_b = \text{id}$. There are two vertices 1, 2. The a -edges form a 2-cycle joining the two vertices, while the b -edges are loops, one at each vertex.

In graph form:

- a : one edge from 1 to 2 and one from 2 to 1,
- b : one loop at 1, one loop at 2.

(2) $\sigma_a = \text{id}, \sigma_b = \tau$. This is the same picture with a and b interchanged:

- a : one loop at each vertex,
- b : a 2-cycle joining the two vertices.

(3) $\sigma_a = \tau, \sigma_b = \tau$. Again there are two vertices 1, 2, and both a and b swap the two sheets:

- a : a 2-cycle joining 1 and 2,
- b : another 2-cycle joining 1 and 2.

Why all degree 2 connected coverings are normal. A connected degree 2 covering corresponds to a subgroup of index 2 in F_2 . Every subgroup of index 2 is normal. Therefore every connected degree 2 covering of $S^1 \vee S^1$ is normal.

So:

There are exactly 3 connected degree 2 coverings, and all of them are normal.

Degree 3. Now we classify transitive pairs in S_3 , up to simultaneous conjugacy. Let

$$c = (123), \quad t = (12).$$

The transitive subgroups of S_3 are $A_3 = \langle c \rangle$ and S_3 . Hence there are two kinds of connected degree 3 coverings: those whose monodromy image is A_3 , and those whose monodromy image is S_3 .

Case A: image A_3 . Then both σ_a, σ_b lie in

$$\{1, c, c^{-1}\},$$

and at least one of them must be a 3-cycle. Up to simultaneous conjugacy, there are four possibilities.

(1) $(\sigma_a, \sigma_b) = (c, \text{id})$. The a -edges form a 3-cycle on the vertices 1, 2, 3, and the b -edges are loops at each vertex.

(2) $(\sigma_a, \sigma_b) = (\text{id}, c)$. This is the same with a and b interchanged: the b -edges form a 3-cycle, and the a -edges are loops.

(3) $(\sigma_a, \sigma_b) = (c, c)$. Both a and b run around the same 3-cycle in the same direction.

(4) $(\sigma_a, \sigma_b) = (c, c^{-1})$. The a -edges and b -edges both form the same 3-cycle, but with opposite orientations.

These four coverings are all normal, because in each case the image is

$$A_3 \cong \mathbb{Z}/3,$$

so the corresponding subgroup has normal quotient of order 3.

Case B: image S_3 . Now the pair (σ_a, σ_b) generates all of S_3 . Up to simultaneous conjugacy, there are three possibilities.

(5) $(\sigma_a, \sigma_b) = (t, c)$. Here

$$\sigma_a = (12), \quad \sigma_b = (123).$$

So:

- a : one vertex carries an a -loop and the other two vertices are joined by an a -edge-pair,
- b : the three vertices form a 3-cycle.

(6) $(\sigma_a, \sigma_b) = (c, t)$. This is the same with a and b interchanged.

(7) $(\sigma_a, \sigma_b) = ((12), (13))$. Here both a and b are transpositions, but different ones. One may describe the graph as follows:

- a : fixes one vertex and swaps the other two,
- b : fixes a different vertex and swaps the other two.

Since (12) and (13) generate S_3 , the action is transitive, so the covering is connected.

These three coverings are not normal. Indeed, if an index 3 subgroup were normal, the quotient would have order 3, so the monodromy image would be cyclic of order 3. Here the image is S_3 , so the subgroup is not normal.

Therefore:

There are exactly 7 connected degree 3 coverings.

Among them:

the first 4 are normal, and the last 3 are not normal.

9 Problem 6 Coverings associated to specific subgroups of $\pi_1(S^1 \vee S^1)$

Problem statement. Consider

$$X = S^1 \vee S^1$$

with basepoint x_0 the wedge point, and

$$\pi_1(X, x_0) = \langle a, b \rangle.$$

Describe covering spaces associated to

- (i) $\langle\langle a \rangle\rangle$, the normal subgroup generated by a ,
- (ii) $\langle a \rangle$, the subgroup generated by a ,
- (iii) the kernel of the homomorphism

$$\phi : \langle a, b \rangle \rightarrow \mathbb{Z}/4$$

given by

$$\phi(a) = [1], \quad \phi(b) = [3] = [-1].$$

What are the fundamental groups of these covering spaces?

Solution.

(i) The covering corresponding to $\langle\langle a \rangle\rangle$

Let

$$H = \langle\langle a \rangle\rangle,$$

the normal closure of a in $F_2 = \langle a, b \rangle$. This is the kernel of the homomorphism

$$F_2 \rightarrow \mathbb{Z}, \quad a \mapsto 0, \quad b \mapsto 1.$$

Hence

$$F_2/H \cong \mathbb{Z}.$$

So the corresponding covering is a connected normal infinite-sheeted covering with deck transformation group $\mathbb{Z}$.

A convenient picture is the following graph. The vertices are indexed by the integers

$$\cdots, -2, -1, 0, 1, 2, \cdots,$$

the b -edges join n to $n + 1$, and at each vertex there is one a -loop. Thus the covering graph is an infinite line in the b -direction with an a -circle attached at every vertex.

Schematically:

$$\cdots \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \cdots$$

with an a -loop at every bullet.

This agrees with the quotient description: a has been killed in the quotient F_2/H , while b survives and acts by deck translations.

The fundamental group of the covering is precisely the subgroup H . Therefore

$$\pi_1(\widetilde{X}) \cong H = \langle\langle a \rangle\rangle.$$

Topologically, this covering graph has infinitely many independent loops, so its fundamental group is a free group of countably infinite rank. A convenient generating family is

$$\{ b^n a b^{-n} : n \in \mathbb{Z} \}.$$

Hence

$$\pi_1(\widetilde{X}) \cong \text{free group of countably infinite rank}.$$

(ii) The covering corresponding to $\langle a \rangle$

Now let

$$H = \langle a \rangle \cong \mathbb{Z}.$$

This subgroup is not normal in F_2 , so the corresponding covering is not normal.

A geometric way to describe this cover is to start with the universal cover of $S^1 \vee S^1$, which is a tree, and then quotient by the action of the subgroup $\langle a \rangle$. The element a acts by translation along one axis in the tree. After quotienting, this axis becomes a single a -loop, while all the remaining branches of the tree descend to infinite trees attached to that loop.

So the covering has:

- one a -cycle,
- infinitely many tree-like branches attached to it,

- no other essential cycles.

Equivalently, the core of the covering graph is a single loop labelled a , and outside that core the graph is a forest.

Since the covering corresponds to H , we have

$$\pi_1(\widetilde{X}) \cong H \cong \mathbb{Z}.$$

Thus the fundamental group of this covering is

$$\pi_1(\widetilde{X}) \cong \mathbb{Z}.$$

(iii) The covering corresponding to $\ker \phi$, where $\phi(a) = 1$ and $\phi(b) = 3 = -1$ in $\mathbb{Z}/4$

Let

$$H = \ker \phi.$$

Since ϕ is surjective onto $\mathbb{Z}/4$, the subgroup H has index 4. Hence the corresponding covering is a connected 4-sheeted normal covering.

Label the four sheets by the elements of $\mathbb{Z}/4$:

$$0, 1, 2, 3.$$

Then the monodromy action is:

$$a : i \mapsto i + 1, \quad b : i \mapsto i - 1,$$

with indices taken modulo 4.

Therefore the covering graph has four vertices 0, 1, 2, 3, and:

- an a -edge from i to $i + 1$,
- a b -edge from i to $i - 1$.

So the a -edges form a 4-cycle in one direction, while the b -edges form a 4-cycle in the opposite direction. As an unlabeled graph, one may think of it as a 4-cycle with parallel a - and b -edges joining consecutive vertices.

Since the covering corresponds to $H = \ker \phi$, we have

$$\pi_1(\widetilde{X}) \cong H.$$

To compute its rank, use the Nielsen–Schreier formula: a subgroup of index d in a free group of rank r has rank

$$1 + d(r - 1).$$

Here $d = 4$ and $r = 2$, so

$$\text{rank}(H) = 1 + 4(2 - 1) = 5.$$

Hence

$$\pi_1(\widetilde{X}) \cong \text{free group of rank } 5.$$

Summary. For the three coverings in this problem we obtain:

(i) For $H = \langle\langle a \rangle\rangle$:

- the covering is an infinite line of b -edges with one a -loop at each vertex,
- it is normal,
- the deck group is $\mathbb{Z}$,
-

$$\pi_1(\widetilde{X}) \cong \text{free group of countably infinite rank.}$$

(ii) For $H = \langle a \rangle$:

- the covering has one a -loop as core, with infinite trees attached,
- it is not normal,
-

$$\pi_1(\widetilde{X}) \cong \mathbb{Z}.$$

(iii) For $H = \ker \phi$:

- the covering is a 4-sheeted normal cover with vertices $0, 1, 2, 3$,
- a acts by $i \mapsto i + 1$, b acts by $i \mapsto i - 1$,
- the deck group is $\mathbb{Z}/4$,
-

$$\pi_1(\widetilde{X}) \cong \text{free group of rank 5.}$$

10 Disjoint unions, wedge sums, and examples

10.1 Disjoint union

Definition. Let X and Y be topological spaces. Their *disjoint union*, denoted

$$X \sqcup Y,$$

is the space obtained by taking a copy of X and a copy of Y and keeping them separate, even if the underlying sets of X and Y are not disjoint.

A standard formal model is

$$X \sqcup Y := (X \times \{0\}) \cup (Y \times \{1\}),$$

with the topology in which a subset $U \subseteq X \sqcup Y$ is open if and only if

$$U \cap (X \times \{0\})$$

is open in $X \times \{0\} \cong X$, and

$$U \cap (Y \times \{1\})$$

is open in $Y \times \{1\} \cong Y$.

Thus the two copies remain completely separate. In particular, if X and Y are nonempty, then $X \sqcup Y$ has at least two components unless one of the spaces is already disconnected in a more complicated way.

Intuition. The disjoint union means:

put the spaces side by side, but do not glue them together.

Examples.

1.

$$[0, 1] \sqcup [0, 1]$$

is two separate copies of the interval. It is not the same space as $[0, 2]$, because $[0, 2]$ is connected whereas $[0, 1] \sqcup [0, 1]$ is disconnected.

2.

$$S^1 \sqcup S^1$$

is just two disconnected circles.

3.

$$S^1 \sqcup S^2$$

is a disconnected space with two connected components, one homeomorphic to a circle and one homeomorphic to a sphere.

4.

$$\{*\} \sqcup S^1$$

is a circle together with one isolated point lying in a separate connected component.

10.2 Wedge sum

Definition. Let (X, x_0) and (Y, y_0) be based spaces. Their *wedge sum*, denoted

$$X \vee Y,$$

is the quotient space

$$X \vee Y := (X \sqcup Y) / (x_0 \sim y_0).$$

So one first forms the disjoint union $X \sqcup Y$, and then identifies the chosen basepoints x_0 and y_0 .

Thus the wedge sum is obtained by gluing the two spaces together at exactly one chosen point.

Intuition. The wedge sum means:

take the two spaces and glue them together at one basepoint.

So the difference from the disjoint union is that the disjoint union keeps the spaces separate, whereas the wedge sum identifies one point of one space with one point of the other.

As a quotient space. The wedge sum is one of the simplest important examples of a quotient construction. One begins with

$$X \sqcup Y$$

and defines an equivalence relation $\sim$ by declaring

$$x_0 \sim y_0$$

and leaving all other points equivalent only to themselves. Then

$$X \vee Y = (X \sqcup Y) / \sim .$$

This construction is fundamental in topology because many spaces are built by gluing along selected points or subspaces.

10.3 Why the choice of basepoints matters

The wedge construction is defined for *based spaces*, so in principle it depends on the chosen points $x_0 \in X$ and $y_0 \in Y$.

For many familiar spaces such as circles and spheres, choosing a different basepoint gives a homeomorphic wedge sum, because these spaces are homogeneous enough that one point looks like any other point. However, in general, the definition really depends on the specified basepoints.

Thus the wedge sum is naturally a construction in the category of based topological spaces.

10.4 Comparing disjoint union and wedge sum

It is useful to place the two constructions side by side.

Disjoint union:

$$X \sqcup Y$$

means:

- no points are identified,
- the two spaces remain separate,
- if X and Y are nonempty, the result is typically disconnected.

Wedge sum:

$$X \vee Y$$

means:

- choose basepoints $x_0 \in X$ and $y_0 \in Y$,

- form $X \sqcup Y$,
- identify exactly the two basepoints,
- if X and Y are connected, the result is connected.

Basic comparison example.

$$S^1 \sqcup S^1$$

is two separate circles, whereas

$$S^1 \vee S^1$$

is the figure eight. So gluing just one point can dramatically change the topology.

10.5 Standard examples of wedge sums

Example 1: $S^1 \vee S^1$. This is the most classical example. It is the space obtained by gluing two circles at one point, and it is homeomorphic to the figure eight.

Its fundamental group is

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z},$$

the free group on two generators.

Example 2: $S^1 \vee S^2$. This is a circle and a sphere glued together at one point. Geometrically, it is a loop with a sphere attached at its basepoint.

Its fundamental group is

$$\pi_1(S^1 \vee S^2) \cong \pi_1(S^1) * \pi_1(S^2) \cong \mathbb{Z} * 1 \cong \mathbb{Z}.$$

So only the circle contributes nontrivial loops.

Example 3: $S^2 \vee S^2$. This is two spheres touching at one point. Since each sphere is simply connected, the wedge is also simply connected:

$$\pi_1(S^2 \vee S^2) \cong 1.$$

However, this space is certainly not homeomorphic to a single S^2 , because at the wedge point the local topology is different: two 2-dimensional pieces meet there.

Example 4: a bouquet of circles. For $m \geq 1$, the wedge

$$\bigvee_{j=1}^m S^1$$

is called a *bouquet of m circles* or a *rose with m petals*. It consists of m loops all meeting at one common point.

Its fundamental group is

$$\pi_1\left(\bigvee_{j=1}^m S^1\right) \cong F_m,$$

the free group on m generators.

Special cases:

- $m = 1$: just S^1 ,
- $m = 2$: the figure eight,
- $m = 3$: three loops meeting at one point.

Example 5: a wedge of spheres of higher dimension. For $m \geq 1$ and $k \geq 2$, the space

$$\bigvee_{j=1}^m S^k$$

is a wedge of m copies of the k -sphere. Since each S^k with $k \geq 2$ is simply connected, the wedge is also simply connected:

$$\pi_1\left(\bigvee_{j=1}^m S^k\right) \cong 1.$$

Thus the wedge behaves very differently for $k = 1$ and for $k \geq 2$.

Example 6: $[0, 1] \vee [0, 1]$. If each interval is based at 0, then gluing the two copies of 0 together gives a V -shaped graph. This is a simple example showing that wedge sums are not restricted to spheres or circles.

Example 7: $S^0 \vee S^1$. Recall that

$$S^0 = \{-1, +1\}$$

consists of two points. Choose one of these as basepoint and glue it to a basepoint of S^1 . Then the resulting space looks like a circle together with one extra point. This example shows that wedge sums involving disconnected spaces can be less geometrically smooth than wedges of connected CW-complexes.

10.6 Wedge sum versus product

It is important not to confuse wedge sum with product.

For example,

$$S^1 \vee S^1$$

is the figure eight, while

$$S^1 \times S^1$$

is the torus. These are very different spaces.

Their fundamental groups already show the difference:

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}, \quad \pi_1(S^1 \times S^1) \cong \mathbb{Z}^2.$$

So wedge sum corresponds to free product on fundamental groups, while product leads to direct product in this case.

10.7 Wedge sum realized as an ordinary union of subspaces

Often a wedge sum can be seen concretely as a union inside a larger space.

For example, inside the torus $S^1 \times S^1$, the subspace

$$(S^1 \times \{*\}) \cup (\{*\} \times S^1)$$

is homeomorphic to

$$S^1 \vee S^1.$$

Indeed, it consists of one horizontal circle and one vertical circle meeting at the point $(*, *)$.

This gives a useful geometric model of the wedge of two circles.

10.8 Fundamental group of a wedge

One of the most important facts about wedge sums is the behaviour of the fundamental group. For suitably nice based spaces, Seifert–van Kampen gives

$$\pi_1(X \vee Y) \cong \pi_1(X) * \pi_1(Y),$$

the free product of the two fundamental groups.

This is one reason wedge sums play such a central role in algebraic topology.

Examples.

1.

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

2.

$$\pi_1(S^1 \vee S^1 \vee S^1) \cong F_3.$$

3.

$$\pi_1(S^1 \vee S^2) \cong \mathbb{Z} * 1 \cong \mathbb{Z}.$$

4.

$$\pi_1(S^2 \vee S^2) \cong 1 * 1 \cong 1.$$

Thus wedge sums of circles produce free groups, while wedges of higher-dimensional spheres remain simply connected.

10.9 Higher-dimensional intuition

The wedge

$$S^n \vee S^n$$

can be thought of as two n -spheres touching at one point.

For $n = 1$, this gives spaces with visible loops and nontrivial free fundamental groups.

For $n \geq 2$, the spheres are simply connected, so the wedge remains simply connected at the level of π_1 , although the space is still topologically distinct from a single sphere.

Thus wedge sums provide a natural way to build spaces by attaching pieces at one point while preserving much of the topology of each piece.

10.10 Finite and infinite wedges

One may also form wedges of more than two spaces:

$$\bigvee_{i \in I} X_i.$$

When the index set I is finite, this behaves in the usual familiar way. For infinite families, one obtains spaces such as

$$\bigvee_{n=1}^{\infty} S^1,$$

the countable bouquet of circles. This is a very important example in topology, although infinite wedge sums can involve subtleties depending on what topological properties one studies.

10.11 Summary

Disjoint union

$$X \sqcup Y$$

means:

keep X and Y separate.

Wedge sum

$$X \vee Y$$

means:

take $X \sqcup Y$ and glue the chosen basepoints together.

Key examples:

$$S^1 \sqcup S^1 = \text{two disconnected circles,}$$

$$S^1 \vee S^1 = \text{the figure eight,}$$

$$S^1 \vee S^2 = \text{a circle and a sphere glued at one point,}$$

$$\bigvee_{j=1}^m S^1 = \text{a bouquet of } m \text{ circles.}$$

The wedge sum is one of the most basic gluing constructions in topology, and it is especially important because it interacts beautifully with the fundamental group through free products.

11 Theory of covering spaces of graphs

Problems 5 and 6 are special cases of one central principle:

covering spaces of $S^1 \vee S^1$ are the geometric realization of subgroups of the free group F_2 .

Since

$$S^1 \vee S^1$$

is the rose with two petals, its fundamental group is

$$\pi_1(S^1 \vee S^1) \cong F_2 = \langle a, b \rangle.$$

Thus every connected covering of $S^1 \vee S^1$ corresponds to a subgroup of F_2 , and every subgroup of F_2 can be realized by such a covering graph.

11.1 Graphs and free groups

A connected graph Γ always has free fundamental group. If Γ has V vertices and E edges, then

$$\text{rank } \pi_1(\Gamma) = E - V + 1.$$

For the rose with two petals

$$R_2 = S^1 \vee S^1,$$

there is one vertex and two loop-edges, so

$$\text{rank } \pi_1(R_2) = 2 - 1 + 1 = 2.$$

Hence

$$\pi_1(R_2) \cong F_2.$$

More generally, if

$$R_n = \bigvee_{j=1}^n S^1,$$

then R_n is the rose with n petals, and

$$\pi_1(R_n) \cong F_n.$$

Thus roses are the standard topological models of free groups.

11.2 Coverings of graphs

A covering of a graph is again a graph. Locally, around each vertex upstairs, the covering must look like a disjoint union of copies of a small neighbourhood of the base vertex.

In the case

$$R_2 = S^1 \vee S^1,$$

the base graph has one vertex and two loop-edges, corresponding to the generators a and b . Therefore, in a covering graph, each vertex must have the correct local pattern of a -edges and b -edges. If one uses oriented labelled edges, then at each vertex there must be exactly one outgoing and one incoming edge of each label.

So a covering of R_2 can be viewed combinatorially as a graph whose edges are labelled by a and b , arranged so that locally it looks exactly like the base rose. This is why the coverings in Problems 5 and 6 can be drawn and classified purely by graph combinatorics.

11.3 Connected coverings and subgroups

A fundamental theorem of covering-space theory says:

Connected coverings of a connected, locally path-connected space X correspond to conjugacy classes of subgroups of $\pi_1(X)$.

Applied to

$$X = S^1 \vee S^1, \quad \pi_1(X) \cong F_2,$$

this becomes:

Connected coverings of $S^1 \vee S^1$ correspond to conjugacy classes of subgroups of F_2 .

If one chooses a basepoint upstairs, then based connected coverings correspond to actual subgroups, not just conjugacy classes.

Thus:

- Problem 5 classifies finite-index subgroups of F_2 of index 2 and 3,
- Problem 6 studies the coverings corresponding to some specific subgroups of F_2 .

11.4 Degree of a covering and index of a subgroup

If a connected covering

$$p : \widetilde{X} \rightarrow X$$

corresponds to a subgroup

$$H \leq \pi_1(X),$$

then the number of sheets of the covering equals the index of the subgroup:

$$\deg(p) = [\pi_1(X) : H].$$

For

$$X = S^1 \vee S^1, \quad \pi_1(X) \cong F_2,$$

this means that a degree n connected covering corresponds exactly to a subgroup of index n in F_2 .

Therefore:

- degree 2 coverings correspond to index 2 subgroups of F_2 ,
- degree 3 coverings correspond to index 3 subgroups of F_2 ,
- infinite-sheeted coverings correspond to infinite-index subgroups.

This is the basic bridge between topology and group theory in Problems 5 and 6.

11.5 Why permutation representations appear

Suppose

$$H \leq F_2$$

has finite index n . Then F_2 acts on the set of left cosets

$$F_2/H.$$

This produces a homomorphism

$$\rho : F_2 \rightarrow S_n,$$

called the *coset action* or *monodromy representation*.

Since

$$F_2 = \langle a, b \rangle,$$

the action is determined by the images of the generators:

$$\sigma_a = \rho(a), \quad \sigma_b = \rho(b).$$

Thus every finite-sheeted covering of $S^1 \vee S^1$ determines a pair of permutations, and conversely every suitable transitive pair of permutations determines a connected covering.

This is exactly why the degree 2 and degree 3 coverings in Problem 5 can be classified by transitive pairs in S_2 and S_3 .

11.6 Why transitivity means connectedness

A finite covering is connected if and only if the monodromy action on the fibre is transitive.

Indeed, if the action is not transitive, then the fibre splits into several orbits. Each orbit corresponds to a separate connected component of the covering. Conversely, if the action is transitive, all sheets lie in the same connected component.

Therefore:

$$\text{connected covering} \quad \Longleftrightarrow \quad \text{transitive monodromy action.}$$

This is the key reason why Problem 5 becomes a classification of transitive permutation representations.

11.7 Normal coverings and normal subgroups

A connected covering is called *normal* (or *regular*, or *Galois*) if the corresponding subgroup is normal.

If

$$H \trianglelefteq \pi_1(X),$$

then the quotient group

$$\pi_1(X)/H$$

acts as the deck transformation group of the covering.

For

$$X = S^1 \vee S^1, \quad \pi_1(X) \cong F_2,$$

this means:

- if $H \trianglelefteq F_2$, then the covering is normal,
- its deck transformation group is

$$F_2/H.$$

This explains several of the results in Problems 5 and 6:

- every index 2 subgroup gives a normal covering,
- every kernel $\ker \phi$ gives a normal covering,
- the normal closure $\langle\langle a \rangle\rangle$ also gives a normal covering.

11.8 Why every index 2 subgroup is normal

A standard group-theoretic fact says:

Every subgroup of index 2 in any group is normal.

Indeed, if $H \leq G$ has index 2, then there are only two left cosets and only two right cosets, so they must coincide.

Therefore every connected degree 2 covering of $S^1 \vee S^1$ is automatically normal.

This explains the answer in Problem 5 for degree 2.

For index 3, the situation changes: some subgroups are normal, some are not. That is why among the degree 3 coverings, some are normal and some are not.

11.9 Schreier graphs and labelled covering graphs

A very useful way to visualize coverings of

$$R_2 = S^1 \vee S^1$$

is through a *Schreier graph*.

One proceeds as follows:

- take one vertex for each sheet, or equivalently for each coset,
- draw an a -edge according to the action of a ,
- draw a b -edge according to the action of b .

Thus, if the fibre is labelled by

$$\{1, \dots, n\},$$

then:

- from vertex i , the a -edge goes to $\sigma_a(i)$,
- from vertex i , the b -edge goes to $\sigma_b(i)$.

This produces a labelled directed graph encoding the covering. If one forgets orientations, one obtains the underlying covering graph.

This is the cleanest combinatorial language for Problems 5 and 6.

11.10 Examples from Problem 5

Degree 2. Since

$$S_2 = \{1, (12)\},$$

transitivity means that at least one of a or b acts by the transposition (12) . This yields exactly three transitive pairs:

$$((12), 1), \quad (1, (12)), \quad ((12), (12)).$$

Each of these gives a connected 2-sheeted covering. Since every index 2 subgroup is normal, all three coverings are normal.

Degree 3. Now one works in S_3 . The transitive subgroups are:

$$A_3 \cong \mathbb{Z}/3, \quad S_3.$$

Thus connected degree 3 coverings split into two families:

1. coverings whose monodromy image is A_3 , which are normal,
2. coverings whose monodromy image is S_3 , which are not normal as degree 3 subgroup coverings.

This is exactly the pattern that appears in Problem 5.

11.11 Coverings corresponding to specific subgroups

Problem 6 studies particular subgroups of F_2 rather than all subgroups of a fixed index. The corresponding coverings are best understood by asking:

- what quotient group survives,
- what the coset graph looks like,
- what loops remain upstairs.

(i) The normal closure $\langle\langle a \rangle\rangle$. This is the smallest normal subgroup containing a . When one quotients by it, the generator a is killed, while b survives. Hence

$$F_2/\langle\langle a \rangle\rangle \cong \mathbb{Z},$$

generated by the image of b .

Geometrically, the covering becomes an infinite line in the b -direction, with one a -loop attached at each vertex. This is a normal infinite-sheeted cover whose deck transformation group is $\mathbb{Z}$.

(ii) The subgroup $\langle a \rangle$. This subgroup is cyclic but not normal in F_2 . Therefore the corresponding covering is not normal.

Its core consists of one essential a -loop, and attached to that loop are infinite trees coming from the rest of the universal cover. Thus only one cyclic direction survives upstairs, while the rest of the graph is tree-like.

So the fundamental group of the covering is

$$\langle a \rangle \cong \mathbb{Z}.$$

(iii) The kernel of $\phi : F_2 \rightarrow \mathbb{Z}/4$. Let

$$H = \ker \phi, \quad \phi(a) = [1], \quad \phi(b) = [-1].$$

Then

$$F_2/H \cong \mathbb{Z}/4,$$

so the covering is connected, 4-sheeted, normal, and has deck group $\mathbb{Z}/4$.

The fibre may be labelled by

$$0, 1, 2, 3 \in \mathbb{Z}/4,$$

and the generators act by

$$a : i \mapsto i + 1, \quad b : i \mapsto i - 1.$$

So both a and b run around the same 4-cycle, but in opposite directions.

This is a clean example of a finite normal covering obtained from a quotient homomorphism.

11.12 Nielsen–Schreier theory

A crucial theorem here is the Nielsen–Schreier theorem:

Every subgroup of a free group is free.

Thus every subgroup of

$$F_2$$

is again a free group.

Moreover, if

$$H \leq F_r$$

has finite index d , then

$$\text{rank}(H) = 1 + d(r - 1).$$

For F_2 , this becomes

$$\text{rank}(H) = 1 + d.$$

Hence:

- an index 2 subgroup has rank 3,
- an index 3 subgroup has rank 4,
- an index 4 subgroup has rank 5.

This explains, for example, the answer in Problem 6(iii): since

$$[F_2 : \ker \phi] = 4,$$

we get

$$\text{rank}(\ker \phi) = 5.$$

So the corresponding covering has fundamental group free of rank 5.

11.13 Core graphs and attached trees

A useful geometric principle is:

Any connected graph is a core graph with trees attached.

The *core* is the smallest subgraph containing all closed loops. Everything outside the core is a forest.

This is especially useful in Problem 6:

- for $\langle a \rangle$, the core is a single loop,
- for $\langle\langle a \rangle\rangle$, the graph has infinitely many loops arranged along a line,
- for $\ker \phi$, the whole finite graph is core.

So the subgroup can often be understood directly from the shape of the core of the covering graph.

11.14 Why Problems 5 and 6 are important

These problems are important because they show three viewpoints at once:

Topological viewpoint. A covering space of a graph is itself a graph lying above the base graph.

Group-theoretic viewpoint. A connected covering corresponds to a subgroup of a free group.

Combinatorial viewpoint. A finite covering can be encoded by permutations of the fibre.

This is one of the cleanest situations in algebraic topology where topology, algebra, and combinatorics become three descriptions of the same structure.

11.15 Summary

For

$$X = S^1 \vee S^1, \quad \pi_1(X) \cong F_2,$$

we have:

- connected coverings correspond to subgroups of F_2 ,
- degree n coverings correspond to index n subgroups,
- normal coverings correspond to normal subgroups,
- finite coverings correspond to transitive permutation representations,
- the covering graph can be drawn as a Schreier graph,
- every subgroup of F_2 is free,

- finite-index subgroup ranks are given by Nielsen–Schreier:

$$\text{rank}(H) = 1 + d(r - 1).$$

Thus Problems 5 and 6 are a concrete study of subgroups of the free group F_2 through labelled graph coverings.

11.16 The identity, nonabelianity, abelian subgroups, and abelianization of F_2

The identity element. Like every group, the free group

$$F_2 = \langle a, b \rangle$$

contains an identity element. It is usually denoted either by

$$e \quad \text{or} \quad 1,$$

and these two symbols mean the same element:

$$e = 1.$$

Thus the elements of F_2 include:

$$1, a, b, a^{-1}, b^{-1}, ab, ba, a^{-1}b^2a, \dots$$

that is, all reduced words in the letters

$$a^{\pm 1}, b^{\pm 1}.$$

The identity element is represented by the empty word, or by any word that reduces completely by cancellation, for example

$$aa^{-1} = 1, \quad abb^{-1}a^{-1} = 1.$$

Why F_2 is not abelian. The free group F_2 is *not* abelian. In particular,

$$ab \neq ba.$$

Indeed, if $ab = ba$, then the commutator

$$[a, b] = aba^{-1}b^{-1}$$

would be equal to the identity. But in F_2 , the word

$$aba^{-1}b^{-1}$$

is already reduced, so it is a nontrivial element. Therefore

$$[a, b] \neq 1,$$

and hence F_2 is nonabelian.

So F_2 is the basic example of a free nonabelian group on two generators.

Abelian subgroups of F_2 . Although F_2 itself is not abelian, it does contain abelian subgroups. However, these are very restricted.

A fundamental fact is:

Every abelian subgroup of a free group is cyclic.

So in F_2 , any abelian subgroup is either:

- the trivial subgroup

$$\{1\},$$

or

- an infinite cyclic subgroup

$$\langle w \rangle \cong \mathbb{Z}$$

generated by some nontrivial word $w \in F_2$.

Typical examples are:

$$\langle a \rangle = \{\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots\} \cong \mathbb{Z},$$

$$\langle b \rangle \cong \mathbb{Z},$$

$$\langle ab \rangle \cong \mathbb{Z}.$$

What does *not* occur inside F_2 is a subgroup such as

$$\mathbb{Z}^2,$$

because that would require two independent commuting elements, and free groups do not contain such abelian subgroups.

Abelianization of F_2 . Although F_2 itself is not abelian, one can force commutativity by passing to the abelianization.

The *commutator subgroup* of F_2 is

$$[F_2, F_2],$$

the subgroup generated by all commutators

$$xyx^{-1}y^{-1} \quad (x, y \in F_2).$$

The *abelianization* is then

$$F_2^{ab} = F_2/[F_2, F_2].$$

For the free group on two generators, the abelianization is

$$F_2^{ab} \cong \mathbb{Z}^2.$$

Indeed, after imposing the relation

$$ab = ba,$$

the generators a and b become independent commuting generators, so the resulting group is the free abelian group on two generators:

$$\mathbb{Z} \oplus \mathbb{Z}.$$

Thus

$$F_2/[F_2, F_2] \cong \mathbb{Z}^2.$$

The abelianization map. There is a natural surjective homomorphism

$$\pi : F_2 \rightarrow \mathbb{Z}^2$$

given by

$$\pi(a) = (1, 0), \quad \pi(b) = (0, 1).$$

This map records how many times a word uses a and b , counting inverses negatively.

For example,

$$\pi(a^2ba^{-1}b^{-3}) = (2, 0) + (0, 1) + (-1, 0) + (0, -3) = (1, -2).$$

Its kernel is exactly the commutator subgroup:

$$\ker(\pi) = [F_2, F_2].$$

So one may write

$$F_2^{ab} \cong \mathbb{Z}^2$$

via this explicit quotient map.

Topological interpretation. Since

$$F_2 \cong \pi_1(S^1 \vee S^1),$$

its abelianization corresponds to the first homology group:

$$H_1(S^1 \vee S^1; \mathbb{Z}) \cong \mathbb{Z}^2.$$

Geometrically, this means:

- $\pi_1(S^1 \vee S^1)$ remembers the full noncommutative loop structure,
- $H_1(S^1 \vee S^1; \mathbb{Z})$ remembers only how many times one goes around each circle, ignoring order.

So:

$$ab \neq ba \quad \text{in } \pi_1,$$

but both determine the same element

$$(1, 1) \in \mathbb{Z}^2$$

after abelianization.

Summary. The free group

$$F_2 = \langle a, b \rangle$$

contains the identity element $1 = e$, is nonabelian, and has only cyclic abelian subgroups. Its abelianization is

$$F_2^{ab} \cong \mathbb{Z}^2.$$

So the key picture is:

F_2 is the basic free nonabelian group on two generators,

while

$\mathbb{Z}^2$ is its abelianized version.

12 The free group F_2 : definition, description, examples, and diagrams

12.1 Definition of the free group F_2

The *free group on two generators* is denoted

$$F_2 = \langle a, b \rangle.$$

This means that F_2 is the group generated by two symbols a and b , with no relations imposed except those required by the group axioms.

Equivalently, F_2 is the most general group generated by two elements. In particular, in F_2 we do *not* impose relations such as

$$ab = ba, \quad a^2 = e, \quad b^3 = e,$$

unless we later pass to a quotient group.

Thus F_2 is free from relations between the generators a and b , and this is why it is called a *free* group.

12.2 Description of the elements of F_2

Elements of F_2 are represented by words in the alphabet

$$a, a^{-1}, b, b^{-1}.$$

A word is called *reduced* if it contains no adjacent inverse pair:

$$aa^{-1}, \quad a^{-1}a, \quad bb^{-1}, \quad b^{-1}b.$$

Every element of F_2 can be represented uniquely by a reduced word.

Examples of words:

$$a, \quad b, \quad ab, \quad ba, \quad a^{-1}b^2a, \quad bab^{-1}a^{-1}.$$

Examples of reduction:

$$abb^{-1}a^{-1} = aa^{-1} = e,$$

so the word $abb^{-1}a^{-1}$ represents the identity.

Also,

$$aba^{-1}a = ab,$$

because the adjacent letters $a^{-1}a$ cancel.

A word such as

$$ab^{-1}ab$$

is already reduced, so it represents a nontrivial element of F_2 .

12.3 Why F_2 is called free

The group F_2 is called free because the generators a and b satisfy no relations other than the formal cancellations that occur in every group.

For example, in F_2 ,

$$ab \neq ba,$$

because commutativity has not been imposed.

Likewise,

$$a^2 \neq e, \quad b^3 \neq e,$$

since no such relations hold in a free group.

Thus F_2 is nonabelian and highly unconstrained. It is the largest possible group generated by two symbols before any additional relations are introduced.

12.4 Universal property of F_2

A fundamental property of F_2 is the following universal property:

If G is any group and $x, y \in G$ are any two elements, then there exists a unique group homomorphism

$$\varphi : F_2 \rightarrow G$$

such that

$$\varphi(a) = x, \quad \varphi(b) = y.$$

This means that F_2 is the most general group generated by two elements. Every group generated by two elements is obtained from F_2 by imposing relations, that is, by taking a quotient of F_2 .

12.5 Topological meaning of F_2

The most important geometric realization of F_2 is:

$$F_2 \cong \pi_1(S^1 \vee S^1).$$

Thus F_2 is the fundamental group of the wedge of two circles, also called the *figure eight* or the *rose with two petals*.

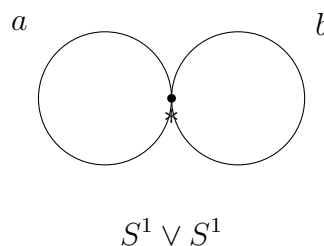
If a denotes a loop going once around the first circle and b denotes a loop going once around the second circle, then words in a and b correspond to paths obtained by traversing those loops in sequence.

For example:

- the word ab means go once around the first loop, then once around the second,
- the word aba^{-1} means go around the first loop, then the second, then backwards around the first.

12.6 Diagram: the rose with two petals

The wedge of two circles is the basic topological model of F_2 .



The basepoint is the common point $*$. One loop gives the generator a , the other gives the generator b . Hence

$$\pi_1(S^1 \vee S^1, *) \cong F_2.$$

12.7 Words as loops

A reduced word in F_2 corresponds to a loop in the rose obtained by following the circles in the order prescribed by the word.

For example, the commutator

$$[a, b] = aba^{-1}b^{-1}$$

means:

1. go around loop a ,

2. then around loop b ,
3. then backwards around a ,
4. then backwards around b .

In the free group F_2 , this element is not equal to the identity. This shows immediately that F_2 is nonabelian.

Indeed, if $ab = ba$, then

$$aba^{-1}b^{-1} = e,$$

but in F_2 this is a nontrivial reduced word.

12.8 The Cayley graph of F_2

Another fundamental geometric object associated to F_2 is its Cayley graph with respect to the generators a and b .

Definition. The vertices of the Cayley graph are the elements of F_2 . From each vertex g , one draws edges to

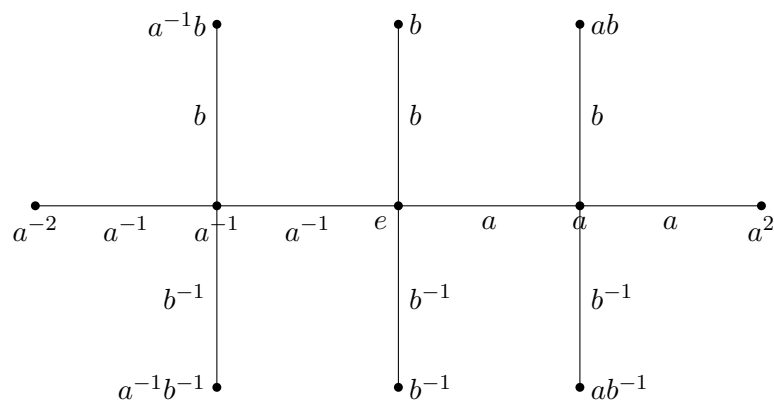
$$ga, \quad ga^{-1}, \quad gb, \quad gb^{-1}.$$

So from every vertex there are four possible moves, corresponding to multiplication by a , a^{-1} , b , and b^{-1} .

Key fact. The Cayley graph of F_2 is an infinite tree.

12.9 Diagram: a fragment of the Cayley graph of F_2

Below is a small fragment of the Cayley graph. The full graph continues infinitely in all directions.



The full Cayley graph branches forever, and every vertex has four incident directed edges, one for each generator or inverse.

12.10 Why the Cayley graph is a tree

The Cayley graph of F_2 has no nontrivial cycles.

Indeed, a cycle in the Cayley graph would correspond to a nonempty reduced word in

$$a^{\pm 1}, b^{\pm 1}$$

that evaluates to the identity. But in a free group, the only word representing the identity is the one that reduces completely to the empty word.

Thus there are no nontrivial cycles, and the Cayley graph is a tree.

This tree is precisely the universal cover of

$$S^1 \vee S^1.$$

12.11 Examples of elements in F_2

Simple elements:

$$e, \quad a, \quad b, \quad a^{-1}, \quad b^{-1}.$$

Reduced nontrivial words:

$$ab, \quad ba, \quad aba, \quad ab^{-1}a, \quad b^2a^{-1}b.$$

Words that reduce:

$$\begin{aligned} aa^{-1} &= e, \\ abb^{-1}a^{-1} &= e, \\ aba^{-1}a &= ab. \end{aligned}$$

Distinct elements: In F_2 , one generally has

$$ab \neq ba, \quad a^2b \neq aba, \quad aba^{-1} \neq b.$$

12.12 Examples of homomorphisms from F_2

Because F_2 is free, any choice of images of a and b in a group G defines a unique homomorphism

$$F_2 \rightarrow G.$$

Example 1: a homomorphism to $\mathbb{Z}$. Define

$$\varphi : F_2 \rightarrow \mathbb{Z}$$

by

$$\varphi(a) = 1, \quad \varphi(b) = 0.$$

Then

$$\varphi(a^2ba^{-1}b^{-3}) = 2 + 0 - 1 + 0 = 1.$$

Example 2: a homomorphism to $\mathbb{Z}/2$. Define

$$\varphi(a) = 1, \quad \varphi(b) = 1.$$

Then both generators map to the nontrivial element of $\mathbb{Z}/2$.

Example 3: a homomorphism to S_3 . Define

$$\varphi(a) = (12), \quad \varphi(b) = (123).$$

Then there is a unique homomorphism

$$F_2 \rightarrow S_3$$

with those images.

This illustrates the universal property of F_2 : every two-generator group receives a homomorphism from F_2 .

12.13 Quotients of F_2

Many familiar groups arise from F_2 by imposing relations.

Example 1: the abelianization. If one imposes the relation

$$ab = ba,$$

then the free group becomes the free abelian group on two generators:

$$F_2/[F_2, F_2] \cong \mathbb{Z}^2.$$

Example 2: killing one generator. If one imposes

$$b = e,$$

then the quotient becomes

$$\langle a \rangle \cong \mathbb{Z}.$$

Example 3: adding torsion relations. If one imposes relations such as

$$a^2 = e, \quad b^3 = e,$$

then one obtains a much smaller quotient group. In this way, F_2 serves as a starting point from which many other groups are constructed.

12.14 Why F_2 is important

The free group F_2 is one of the most important basic objects in algebra and topology because:

- it is the simplest nonabelian free group,
- every group generated by two elements is a quotient of F_2 ,
- it is the fundamental group of the figure eight,
- its Cayley graph is a tree,
- coverings of $S^1 \vee S^1$ correspond to subgroups of F_2 .

So F_2 lies exactly at the intersection of:

- combinatorial group theory,
- graph theory,
- covering-space theory,
- algebraic topology.

12.15 Summary

The free group on two generators is

$$F_2 = \langle a, b \rangle.$$

Its elements are reduced words in

$$a^{\pm 1}, b^{\pm 1}.$$

It has no defining relations except cancellation of inverse pairs. It is nonabelian and infinite.

Geometrically,

$$F_2 \cong \pi_1(S^1 \vee S^1),$$

where $S^1 \vee S^1$ is the rose with two petals. Combinatorially, the Cayley graph of F_2 is an infinite tree, which is the universal cover of $S^1 \vee S^1$.

Thus F_2 is the basic example linking words, groups, graphs, and covering spaces.

13 Fundamental groups: definition, theory, examples, and diagrams

13.1 The basic idea

The fundamental group measures how loops in a space can or cannot be deformed into one another.

Very roughly:

- if every loop can be shrunk to a point, then the space has no nontrivial one-dimensional hole;
- if some loop cannot be shrunk to a point, then the space has nontrivial fundamental group.

So the fundamental group is a topological invariant that detects the basic loop-structure of a space.

13.2 Paths and loops

Let X be a topological space.

A *path* in X is a continuous map

$$\gamma : [0, 1] \rightarrow X.$$

The point

$$\gamma(0)$$

is the starting point of the path, and

$$\gamma(1)$$

is the endpoint.

A *loop* based at a point $x_0 \in X$ is a path

$$\gamma : [0, 1] \rightarrow X$$

such that

$$\gamma(0) = \gamma(1) = x_0.$$

So a loop starts and ends at the same point.

13.3 Homotopy of loops

Let $\gamma, \delta : [0, 1] \rightarrow X$ be loops based at x_0 . We say that γ and δ are *homotopic relative to endpoints* if there exists a continuous map

$$H : [0, 1] \times [0, 1] \rightarrow X$$

such that

$$H(t, 0) = \gamma(t), \quad H(t, 1) = \delta(t)$$

for all $t \in [0, 1]$, and

$$H(0, s) = H(1, s) = x_0$$

for all $s \in [0, 1]$.

Thus H is a continuous deformation from γ to δ , during which the basepoint remains fixed at all times.

13.4 Definition of the fundamental group

Fix a basepoint $x_0 \in X$. The *fundamental group* of X at x_0 , denoted

$$\pi_1(X, x_0),$$

is the set of homotopy classes of loops based at x_0 , with multiplication defined by concatenation of loops.

If $[\gamma]$ denotes the homotopy class of a loop γ , then the product is

$$[\gamma] \cdot [\delta] = [\gamma * \delta],$$

where $\gamma * \delta$ means: first go around γ , then go around δ .

So $\pi_1(X, x_0)$ is not merely a set; it is a group.

13.5 The group operation

If γ, δ are loops at x_0 , their concatenation is defined by

$$(\gamma * \delta)(t) = \begin{cases} \gamma(2t), & 0 \leq t \leq \frac{1}{2}, \\ \delta(2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The identity element of $\pi_1(X, x_0)$ is the class of the constant loop

$$c_{x_0}(t) = x_0.$$

The inverse of a loop γ is the loop traversed backwards:

$$\gamma^{-1}(t) = \gamma(1 - t).$$

Hence in the fundamental group,

$$[\gamma]^{-1} = [\gamma^{-1}].$$

Associativity holds up to homotopy, which is why the multiplication becomes genuinely associative on homotopy classes.

13.6 Why the basepoint matters

The fundamental group is written

$$\pi_1(X, x_0)$$

because loops are based at a chosen point x_0 .

In general, different basepoints can give different groups, but if X is path-connected, then any path from x_0 to x_1 determines an isomorphism

$$\pi_1(X, x_0) \cong \pi_1(X, x_1).$$

So in a path-connected space one often speaks informally of *the* fundamental group of X , although strictly speaking one should still specify the basepoint.

13.7 Simply connected spaces

A space X is called *simply connected* if:

1. X is path-connected, and
2. every loop in X can be shrunk to a point.

Equivalently,

$$\pi_1(X, x_0) = 1$$

for some, and hence every, basepoint $x_0 \in X$.

Examples of simply connected spaces include:

$$\mathbb{R}^n, \quad D^2, \quad S^n \text{ for } n \geq 2.$$

13.8 First geometric intuition

The fundamental group detects one-dimensional holes.

- A disk has no hole, so every loop shrinks.
- A circle has one essential hole, so loops can wind around it.
- A figure eight has two basic loop directions, with no relation forcing them to commute.
- A torus has two basic loop directions that do commute.

This already suggests that the fundamental group can distinguish spaces that may look superficially similar.

13.9 Example: a point

If $X = \{*\}$, then every loop is constant. Hence

$$\pi_1(\{*\}) = 1.$$

13.10 Example: an interval and contractible spaces

If $X = [0, 1]$, then every loop can be contracted to a point. Thus

$$\pi_1([0, 1]) = 1.$$

More generally, every contractible space has trivial fundamental group.

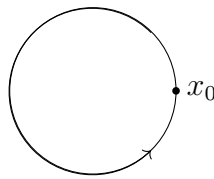
13.11 Example: the circle S^1

One of the most important facts in algebraic topology is

$$\pi_1(S^1) \cong \mathbb{Z}.$$

The integer records how many times a loop winds around the circle, counting direction:

- +1: once counterclockwise,
- +2: twice counterclockwise,
- -1: once clockwise,
- 0: contractible.



$$\pi_1(S^1, x_0) \cong \mathbb{Z}$$

13.12 Example: the punctured plane

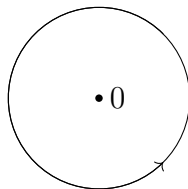
The space

$$\mathbb{R}^2 \setminus \{0\}$$

also has fundamental group

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}.$$

Indeed, the missing point creates one hole, and loops are classified by how many times they wind around that hole. This space deformation retracts onto a circle, so it has the same fundamental group as S^1 .



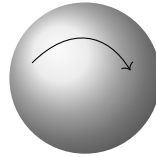
$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}$$

13.13 Example: the sphere S^2

The 2-sphere satisfies

$$\pi_1(S^2) = 1.$$

Although the sphere is a curved surface in $\mathbb{R}^3$, every loop on the sphere can still be deformed to a point. Thus S^2 is simply connected.



$$\pi_1(S^2) = 1$$

13.14 Example: the wedge $S^1 \vee S^1$

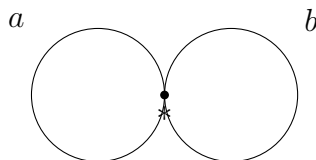
The wedge of two circles has

$$\pi_1(S^1 \vee S^1) \cong F_2 = \langle a, b \rangle,$$

the free group on two generators.

One loop gives the generator a , and the other gives the generator b . No commutativity relation is imposed, so generally

$$ab \neq ba.$$



$$\pi_1(S^1 \vee S^1, *) \cong F_2$$

13.15 Example: the torus

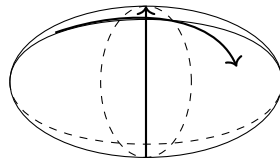
For the torus

$$T^2 = S^1 \times S^1,$$

one has

$$\pi_1(T^2) \cong \mathbb{Z}^2.$$

There are two basic loop directions: one goes around the torus one way, and the other goes around it the other way. These two generators commute, so the group is abelian.



$$\pi_1(T^2) \cong \mathbb{Z}^2$$

Thus

$$\pi_1(S^1 \vee S^1) \cong F_2 \quad \text{but} \quad \pi_1(S^1 \times S^1) \cong \mathbb{Z}^2,$$

and this difference is one of the most important contrasts in basic algebraic topology.

13.16 Example: higher-dimensional spheres

For every $n \geq 2$,

$$\pi_1(S^n) = 1.$$

So among spheres, only the circle S^1 has nontrivial fundamental group.

13.17 Example: real projective space

For $n \geq 2$,

$$\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2.$$

This is a standard example of a space with nontrivial finite fundamental group.

13.18 Fundamental group of products

For suitably nice spaces,

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

For example,

$$\pi_1(S^1 \times S^1) \cong \pi_1(S^1) \times \pi_1(S^1) \cong \mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^2.$$

This explains why the torus has fundamental group $\mathbb{Z}^2$.

13.19 Fundamental group of wedges

For suitably nice based spaces, Seifert–van Kampen gives

$$\pi_1(X \vee Y) \cong \pi_1(X) * \pi_1(Y),$$

the free product of the two groups.

Examples:

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F_2,$$

$$\pi_1(S^1 \vee S^2) \cong \mathbb{Z} * 1 \cong \mathbb{Z},$$

$$\pi_1(S^2 \vee S^2) \cong 1 * 1 \cong 1.$$

So wedge sums naturally lead to free products.

13.20 Induced maps on fundamental groups

If

$$f : (X, x_0) \rightarrow (Y, y_0)$$

is a continuous map with

$$f(x_0) = y_0,$$

then every loop in X based at x_0 is sent to a loop in Y based at y_0 . Thus f induces a group homomorphism

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0).$$

This is called the induced homomorphism on fundamental groups.

So the fundamental group is not just a group attached to a space; it is part of a functorial construction.

13.21 Homotopy invariance

If two spaces are homotopy equivalent, then their fundamental groups are isomorphic.

Thus, if

$$X \simeq Y,$$

then

$$\pi_1(X) \cong \pi_1(Y).$$

This is why deformation retractions are so useful in computing fundamental groups.

For example,

$$\mathbb{R}^2 \setminus \{0\} \simeq S^1,$$

hence

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \pi_1(S^1) \cong \mathbb{Z}.$$

13.22 Covering spaces and the fundamental group

One of the deepest and most useful connections in topology is the relation between the fundamental group and covering spaces.

If

$$p : \widetilde{X} \rightarrow X$$

is the universal cover of a sufficiently nice space X , then:

- $\widetilde{X}$ is simply connected,
- loops downstairs lift to paths upstairs,
- deck transformations of $\widetilde{X}$ encode the fundamental group of X .

Typical examples:

$$\mathbb{R} \rightarrow S^1, \quad \mathbb{R}^2 \rightarrow T^2, \quad \text{an infinite tree} \rightarrow S^1 \vee S^1.$$

So the fundamental group controls how a space “unwraps” into its universal cover.

13.23 The Seifert–van Kampen theorem

The Seifert–van Kampen theorem is one of the main tools for computing fundamental groups.

Very roughly, if

$$X = U \cup V$$

with U , V , and $U \cap V$ sufficiently nice and path-connected, then $\pi_1(X)$ can be computed from

$$\pi_1(U), \quad \pi_1(V), \quad \pi_1(U \cap V).$$

In general, this gives an amalgamated free product. In special cases:

- for wedge sums, it gives a free product,
- for many glued spaces, it produces a group presentation for $\pi_1(X)$.

For example, it gives

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}.$$

13.24 Fundamental group versus first homology

The fundamental group remembers the order in which loops are traversed, so it can be nonabelian.

For example,

$$\pi_1(S^1 \vee S^1) \cong F_2,$$

and in F_2 ,

$$ab \neq ba.$$

By contrast, first homology forgets order and becomes abelian:

$$H_1(S^1 \vee S^1; \mathbb{Z}) \cong \mathbb{Z}^2.$$

So the fundamental group is finer and more delicate than first homology.

13.25 A compact visual summary

Circle:

$$\pi_1(S^1) \cong \mathbb{Z}$$

Figure eight:

$$\pi_1(S^1 \vee S^1) \cong F_2$$

Torus:

$$\pi_1(T^2) \cong \mathbb{Z}^2$$

Sphere:

$$\pi_1(S^2) = 1$$

These four examples already capture the main possibilities:

- cyclic,
- free nonabelian,
- free abelian of rank 2,
- trivial.

13.26 Summary

The fundamental group

$$\pi_1(X, x_0)$$

is the group of loops based at x_0 , modulo homotopy.

It measures the one-dimensional hole structure of a space.

Key examples are:

$$\pi_1(\{*\}) = 1, \quad \pi_1(S^1) = \mathbb{Z}, \quad \pi_1(S^1 \vee S^1) = F_2, \quad \pi_1(S^1 \times S^1) = \mathbb{Z}^2, \quad \pi_1(S^n) = 1 \text{ for } n \geq 2.$$

The main ideas and tools surrounding the fundamental group are:

- loops and homotopy,
- concatenation of loops,

- homotopy invariance,
- induced homomorphisms,
- products and wedge sums,
- universal covers and deck transformations,
- the Seifert–van Kampen theorem.

Thus the fundamental group is one of the central invariants of algebraic topology, connecting geometric intuition, group theory, and covering-space theory.